Notas de Combinatória

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October 27, 2026

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0 Informações importantes

Devo guardar notas de aula e exercícios interessantes do curso. Por enquanto nenhuma notação muito esotérica apareceu, se sim devo listá-las aqui. O objetivo dessas notas é serem breves, de forma que é possível recuperar formalmente as provas dos teoremas enunciados. Não serei estritamente rigoroso, o que pode levar a alguns erros. A linguagem pode variar intermintentemente entre português e inglês.

O código fonte desse arquivo pode ser encontrado em https://github.com/hnrq104/comb2/tree/main.

1 Extremal Set Theory

Lemma 1.1. (LYMB) Let $\mathcal{A} \subset \mathcal{P}([n])$ be an anti-chain, then

$$\sum_{A \in \mathcal{A}} \binom{n}{|A|}^{-1} \leqslant 1$$

Proof. (Sketch) Take a random permutation and consider the indicator random variables X_A that the first $|A|$ elements form A. We find that $\mathbb{E} \sum X_A = \sum_{A \in \mathcal{A}} \binom{n}{|A|}^{-1}$, but $\sum X_A \leqslant 1$. □

Corollary 1.2. (Sperner) If $\mathcal{A}$ is an anti-chain, $|\mathcal{A}| \leqslant \binom{n}{n/2}$.

Corollary 1.3. (Littlewood-Offord) Given $a_1, a_2, \dots a_n \in \mathbb{R} \setminus \{0\}$. Consider $x_1, \dots x_n \in \{-1, 1\}$ u.i.i.d r.v's, then $\mathbb{P}(\sum a_i x_i = 0) \leqslant 2^{-n}\binom{n}{n/2}$.

Proof. By simmetry of the x_i, we may assume that all a_i are positive. Now consider the collection $\mathcal{A} := \{A \subset [n] : \sum_{i\in A} a_i - \sum_{i\notin A} a_i = 0\}$. $\mathcal{A}$ is an anti-chain. □

Theorem 1.4. (Erdos-Ko-Rado, Katona) Let $\mathcal{A} \subset \binom{[n]}{k}$ be an intersecting family with $2k < n$, then $|\mathcal{A}| \leqslant \binom{n-1}{k-1}$.

Proof. (Sketch) Take a random circular permutation of $[n]$, let X_A be the indicator of A appearing as an interval on the circle. We find that $\sum X_A \leqslant k$ - partition the intervals that intersect A and use pigeonhole. Calculating expectation and rearranging gives the result. □

Theorem 1.5. (Exercise ERK) There is $n_0(k, l)$ such that if $n \geqslant n_0$ and $\mathcal{A} \subset \binom{[n]}{k}$ is an l-intersecting family, then $|\mathcal{A}| \leqslant \binom{n-l}{k-l}$. Hints:

1. Show that there exists $L \subset [n]$, $|L| = l$ such that $L = A \cap B$ for $A, B \in \mathcal{A}$.
2. Show that either $\mathcal{A} \subset \partial L := \{A \in \binom{[n]}{k} : L \subset A\}$ or $|\mathcal{A}| = O(n^{k-l-1})$.

Proof. For the first step, assume $\mathcal{A}$ is not empty and for every $A, B \in \mathcal{A}$, $|A \cap B| \geqslant l + 1$. Choose any element C from $\mathcal{A}$ and notice that every set in the family must intersect at least with $l + 1$ elements from C, so the number of sets in $\mathcal{A}$ is bounded by

$$|\mathcal{A}| \leqslant \binom{k}{l+1}\binom{n-l-1}{k-l-1} = O(n^{k-l-1}).$$

So we may assume that there are A, B, L as in the hint. Now, either $\mathcal{A} \subset \partial L$ and we are done, or there exists $C \in \mathcal{A}$ such that $L \not\subset C$. How many such C can exists? Because $A \cap B = L$, $L \not\subset C$, $|A \cap C| \geqslant l$ and $|B \cap C| \geqslant l$ we find that

$$|C \cap (A \cup B)| = |C \cap (A \cap B)| + |C \cap (A \setminus B)| + |C \cap (B \setminus A)| \geqslant l + 1$$

where we used the fact that $l \leqslant |C \cap A| = |C \cap L| + |C \cap (A \setminus B)|$ and so $|C \cap (A \setminus B)| \geqslant 1$. So the number of such C is at most $\binom{|A\cup B|}{l+1}\binom{n-|A\cup B|}{k-l-1} = O(n^{k-l-1})$. And now, fix any such C, how many elements of $\mathcal{A}$ may contain L? Because they all intersect in C an element outside of L, we have at most

$$|C \setminus L|\binom{n-l-1}{k-l-1} = O(n^{k-l-1}).$$

such elements, which completes the proof. □

I really liked Rob's motivation for continuing to study these extremal problems.

(a) An explicit construction that gives $R(k) \geqslant k^C$ for any fixed C!

(b) The chromatic number of $\mathbb{R}^n$. We define $\chi(\mathbb{R}^n)$ as the minimal number of colors k such that there exists a function $f : \mathbb{R}^n \to [k]$ satisfying that for every $x, y \in \mathbb{R}^n$ with $|x - y| = 1$, we have that $f(x) \neq f(y)$. For example, we know by taking a simplex on $\mathbb{R}^n$ that $\chi(\mathbb{R}^n) \geqslant n + 1$. It is possible to show (by considering a tesselation by cubes with diameter less than 1) that the number is well defined for any dimension. We will use extremal stuff to prove some order of growth in n of this parameter.

(c) (Borsuk's Conjecture) Show that for every set $X \subset \mathbb{R}^n$ it is possible to decompose it in $n + 1$ sets $X_1, \dots X_{n+1}$ s.t. $\operatorname{diam}(X_i) < X$.

Now lets go back to some theorems. The next few ones are nice applications of linear algebra.

Definition 1.6. Given $L \subset [n]$, we say a family $\mathcal{A} \subset \mathcal{P}(n)$ is L-intersecting if for every $A \neq B \in \mathcal{A}$ we have that $|A \cap B| \in L$.

Theorem 1.7. (Modular Frankl-Wilson) Let $\mathcal{A} \subset \mathcal{P}(n)$ and let $L \subseteq \mathbb{Z}_p$ with $|L| = s$. If $\mathcal{A}$ is L-intersecting and for every $A \in \mathcal{A}$, $|A| \pmod p \notin L$, then

$$|\mathcal{A}| \leqslant \sum_{i=0}^{s} \binom{n}{i}$$

Proof. Consider the L.I set

$$f_A(X) = \prod_{l \in L} (X \cdot \mathbf{1}_A - l)$$

in $\mathbb{F}_p[x_1, x_2 \dots x_n]$ and calculate the dimension. □

Theorem 1.8. (Frankl-Wilson) Let $\mathcal{A} \subset \binom{n}{k}$ be L intersecting, then $|\mathcal{A}| \leqslant \binom{n}{\leqslant |L|}$.

Proof. Consider $f_A = \prod_{l \in L} (X \cdot \mathbf{1}_A - l) \in \mathbb{R}[x_1, \dots, x_n]$, this always has degree less than $|L|$ and all monomials (after correcting the exponents) have degree at most $|L|$. So, by the dimension bound (of vectors being L.I), we know that $|\mathcal{A}| \leqslant \binom{n}{\leqslant |L|}$. □

Now let's sketch some applications.

Theorem 1.9. There is a explicit coloring that shows $R(m) \geqslant m^C$ for any constant C. (In our setup) There exists a sequence of graphs on N vertices with no clique or independent set of size $N^{o(1)}$.

Proof. The idea is to consider a graph, such that cliques and independent sets correspond to L intersecting families. So let's fix our $V(K_N) = \binom{[n]}{k}$. And consider a coloring of $E[K_N]$ by making

$$c(A, B) = \begin{cases} \text{R} & \text{if} \quad |A \cap B| \in L \\ \text{G} & \text{if} \quad |A \cap B| \notin L \end{cases}$$

where $L \subset [k-1]$. Now, what's the maximum size of a red clique? We know by [1.8], that it is at most by $\binom{n}{\leqslant |L|}$. So if L is relatively small, we can expect to win something. What about green cliques? Then we turn to the modular theorem [1.7]. If we suppose $k = p^2 - 1$ and let $L = \mathbb{Z}_p \setminus \{-1\}$. We would have that each green clique satisfies the hypothesis of 1.7, and so has size at most $\binom{n}{\leqslant p-1}$.

So thats the idea, we take $L = \{p-1, 2p-1, \dots p(p-1)-1\}$ so our coloring has no red clique of size $\binom{n}{\leqslant p-1}$, and no green clique of size $\binom{n}{\leqslant p-1}$.

Suffices to show that if $N = \binom{n}{k}$, then $\binom{n}{\leqslant \sqrt{k}} = N^{o(1)}$. □

Theorem 1.10. $\chi(\mathbb{R}^n) \geqslant \exp(cn)$

Proof. The proof follows similarly, we somehow want to find a finite embeddable graph in $\mathbb{R}^n$ with small independence number. Then $\chi(\mathbb{R}^n) \geqslant \chi(G) \geqslant |V(G)|/\alpha(G)$. So, let's look into our setting again of $V(G) = \binom{[n]}{k}$, and to embedd stuff in $\mathbb{R}^n$ we will consider the indicator functions k-sets in $\mathbb{R}^n$ (which are vectors of n dimensions).

Now we want to forbid a distance between these vectors (which will make our sets independent). So let's forbid a specific distance (which we can always scale to be the right thing) and say

$$E(G) = \{(A, B) : |A \cap B| \neq t\}$$

For some $t \in \mathbb{Z}$. Here as we don't really care about the clique number (only the independence), so we need to bound the size of a independent set. So in order to get the strongest bound, we use theorem [1.7] and suppose $k = 2p - 1$ for some p and $L = t \pmod p$. We get $\alpha(G) \leqslant \binom{n}{1}$. So by taking k very big (say $n/2$) we get the result.

□

Theorem 1.11. (Disproof of Borsuk's Conjecture) $\mathrm{Bor}(\mathbb{R}^n) \geqslant (1+\varepsilon)^{\sqrt{n}}$.

Proof. The proof can be found in Rob's book from 2010 with Imbuzeiro, but follows the same paradigm. □

2 Ramsey Theory

Just for sake of completeness, we define

$$R(k, l) = \min\{n : K_k \not\subset G \subset K_n \Rightarrow \alpha(G) \geqslant l\}.$$

In coloring terms, it's the smallest n such that any red and blue coloring of $E(K_n)$ contains a red K_k or a blue K_l.

2.1 Bounds on R(3,k)

We have two very simple bounds that are easy to prove,

$$\left(\frac{k}{\log(k)}\right)^{3/2} \leqslant R(3,k) \leqslant \binom{k+1}{2}.$$

The upperbound follows easily from the Erdös-Szekeres proof.

Theorem 2.1. $R(3,k) \leqslant \binom{k+1}{2}$

Proof. Let G on n-vtx be a triangle-free with no independet set of size k, we know every neighborhood is an independent set so $\Delta(G) < k$. Now we greedily get one vtx at a time removing their neighbors. Pick v_1 and $N(v_1)$ with $|N(v_1)| \leqslant k-1$, then pick v_2, notice that the neighborhood of v_2 (after removing v_1 and its neighbors) has size at most $k-2$, if not $\{v_1\} \cup N(v_2)$ would be a independent set of size k. We proceed like this for $v_1, v_2, \dots v_t$, where each step we discard at most $k-(i-1)$ vtxs. Because we can't do this process k times, for some $i < k$ we must fail (meaning there are no vertex left), so $n = |V(G)| \leqslant \sum_{j=0}^{k-2} k - j \leqslant \binom{k+1}{2} - 1$. □

The lowerbound is a beautiful application of the probabilistic method. The main idea is to choose $G(n,p)$ as to have few triangles (linearly small in n) and relatively low independence number.

Theorem 2.2.

$$R(3,k) \geqslant \left(\frac{k}{\log(k)}\right)^{3/2}$$

We need a **very important** lemma first.

Lemma 2.3. With high probability, the independence number of $G(n,p) \leqslant 2\log(n)/p$. That is

$$\mathbb{P}(\alpha(G(n,p)) \geqslant 2\log(n)/p + 1) = 1 - o_{(n,p(n))}(1)$$

where p can be taken polynomially small in n.

Proof.

$$\begin{aligned}
\mathbb{P}(\alpha(G(n,p)) \geqslant k) &\leqslant \mathbb{E}[\#\text{independent k sets} \in G(n,p)] \\
&= \binom{n}{k}(1-p)^{\binom{k}{2}} \leqslant \left(\frac{en}{k}\right)^k \cdot e^{-pk(k-1)/2} \\
&= \left(\frac{en}{e^{p(k-1)/2}k}\right)^k
\end{aligned}$$

Taking $k \geqslant 2\log(n)/p + 1$ we get that

$$\mathbb{P}(\alpha(G(n,p)) \geqslant k) \leqslant \left(\frac{ep}{2\log(n)}\right)^{2\log(n)/p} \to 0$$

very fast even! (faster than any polynomial). □

The strategy will be to remove triangles from $G(n,p)$ by removing one vertex from each. This cannot increase the independence number, so we hope the resulting graph still has many vertices to yield a Ramsey construction.

So let's see what this alteration may give us.

Proof. (Of the Theorem) We choose p small such that the number of triangles is no larger than a small linear in n (say $n/2$).

$$\mathbb{E}[\#K_3 \subset G(n,p)] = \binom{n}{3}p^3 = (1+o(1))\frac{(pn)^3}{6}$$

We want

$$\mathbb{P}(\#K_3 \in G(n,p) \geqslant n/2) < (1+o(1))\frac{p^3n^2}{3} < 1/2$$

so taking $p = \varepsilon n^{-2/3}$ for some small ε we get the inequality. But what does this p gives in independence number? We have w.h.p (say bigger than one half)

$$\alpha(G(n,p)) \leqslant 2\varepsilon^{-1}\log(n)n^{2/3}.$$

So $\mathbb{P}(K_3(G(n,p)) < n/2) > 1/2$ and $\mathbb{P}(\alpha(G(n,p) < C\log(n)n^{2/3})) > 1/2$, meaning there exists G satisfying both statements. Remove one vtx from each triangle in G, we are left with a triangle-free graph G' with at least $n/2$ vtxs and $\alpha(G) < C\log(n)n^{2/3}$. Setting $k = C\log(n)n^{2/3}$, we get that

$$V(G') = n/2 \geqslant c\left(\frac{k}{\log(k)}\right)^{3/2}$$

□

2.1.1 Krivilevich lowerbound

Now for a better (and astounishingly beautiful) bound.

Theorem 2.4.

$$R(3,k) \geqslant \left(\frac{k}{\log(k)}\right)^2.$$

The idea of this bound is removing edges. Maybe we can kill the triangles by removing few edges and not increase the independence number too much. For this we will need a beautiful lemma.

Lemma 2.5. (Erdös-Tetali) Let $\{A_1, A_2, \dots A_m\}$ be a family of events in some probability space (Ω, P). Let $\mathcal{E}_t$ be the event that at least t different independent A_i happen. We have that

$$\mathbb{P}(\mathcal{E}_t) \leqslant \frac{1}{t!}\left(\sum_{i=1}^{m} A_i\right)^t.$$

Proof. We get this one from just opening the terms

$$\mathbb{P}(\mathcal{E}_t) = \mathbb{P}\left(\bigcup_{\substack{i_1<\cdots<i_t \\ A_{i_j} \not\sim A_{i_k}}} \bigcap_{j=1}^{t} A_{i_j}\right) \leqslant \sum_{\substack{i_1<\cdots<i_t \\ A_{i_j} \not\sim A_{i_k}}} \prod_{j=1}^{t} \mathbb{P}(A_{i_j}) \leqslant \sum_{i_1<\cdots<i_t} \prod_{j=1}^{t} \mathbb{P}(A_{i_j}) \leqslant \frac{1}{t!}\left(\sum_{i=1}^{m} A_i\right)^t.$$

□

The second ingredient is noting that if you're a bit bigger (say a constant times) than the independent size, you already have many edges (at least half of what is expected).

Lemma 2.6. Let $k \geqslant C\log(n)/p$, then

$$\mathbb{P}\left(\exists \text{ k-set } S, e(G(n,p)[S]) < p\binom{k}{2}/2\right) = o(1)$$

for some $C > 0$.

Proof. Let $\mathcal{E}$ be the event that there exists a k-set S with $e(G(n,p)[S]) < p\binom{k}{2}/2$. We will do a union bound and use Hoeffding (or Chernoff) to bound its probability.

Let S be any k-set. Note that (letting $B(n,p)$ be the binomial with parameters n and p) we have

$$e(G(n,p)[S]) \equiv B\left(\binom{k}{2}, p\right).$$

So by Hoeffding and its bounds on the binomial [A.5] (need to add proofs for these inequalities),

$$\mathbb{P}\left(e(G(n,p)) < \frac{p}{2}\binom{k}{2}\right) = \mathbb{P}\left(\left|B\left(\binom{k}{2}, p\right) - p\binom{k}{2}\right| \geqslant \frac{p}{2}\binom{k}{2}\right) \leqslant e^{-cp\binom{k}{2}}$$

for some $c > 0$.

Now, doing the union bound over k-sets we find that (taking $C > 3c^{-1}$)

$$\mathbb{P}(\mathcal{E}) \leqslant \binom{n}{k} e^{-cp\binom{k}{2}} \leqslant \left(\frac{en}{e^{cp(k-1)/2}k}\right)^k \leqslant \left(\frac{e}{k}\right)^k \to 0.$$

□

The idea to prove theorem 2.4 is to remove a maximal collection of edge-disjoint triangles from $G(n,p)$, (notice that this makes it triangle free). We use lemma 2.5 to show that this procedure won't remove too many edges from any set S with size $C\log(n)/p$.

Proof. (Of Theorem 2.4) Let S be a set of size $k = C\log(n)/p$, for some p we will determine later. Let $\{T_1, \dots T_m\}$ be the set of all triangles that intersect with at least a pair in S. Note that $m \leqslant n\binom{|S|}{2}$. Let $\mathcal{E}_S$ be the event that a edge disjoint set of these triangles with size bigger or equal to $\frac{p}{6}\binom{k}{2}$ are in $G(n,p)$. By [2.5], we can bound $\mathbb{P}(\mathcal{E}_S)$,

$$\mathbb{P}(\mathcal{E}_S) \leqslant \frac{1}{[\frac{p}{6}\binom{k}{2}]!}\left(\sum_{i=1}^{m}\mathbb{P}(T_i \in G(n,p))\right)^t \leqslant \left(\frac{24e}{pk^2}\right)^{pk^2/24}(p^3nk^2)^{pk^2/12}$$

Replacing k in the equation we get (and putting $C > 24e$)

$$\mathbb{P}(\mathcal{E}_S) \leqslant \left(\frac{24ep}{C\log^2(n)}\right)^{\log(n)^2/(24p)}(pn\log(n)^2)^{\log(n)^2/(12p)} \leqslant (p^3n^2\log(n)^2)^{\log(n)^2/(24p)}$$

So, if we take $p = \varepsilon n^{-1/2}$ (might try to optimize more later), we already find that

$$\mathbb{P}(\mathcal{E}_S) \leqslant \left(\frac{\log(n)}{n^{1/2}}\right)^{n^{1/2}\log^2(n)} \ll \left(\frac{C\log(n)}{n^{1/2}}\right)^{C\log(n)n^{1/2}} \leqslant \binom{n}{k}^{-1}$$

So $\mathcal{E}_S$ does not happen for any S with high probability. As w.h.p S of size k have more than $\frac{p}{2}\binom{k}{2}$ edges, and no maximal disjoint triangle family intersect too much with S. We can remove a maximal edge disjoint triangle family and not make any set of size k independent. This yields a triangle free graph on n vtxs with independence number less than $C\log(n)n^{1/2}$, so

$$R(3,k) \geqslant c\left(\frac{k}{\log k}\right)^2.$$

□

2.1.2 AKSz Lemma

Now let's find an upperbound of the right order.

Theorem 2.7. (Ajtai, Komlós, Szemerédi)

$$R(3,k) \leqslant C\frac{k^2}{\log k}$$

The theorem will be immediately implied by the following lemma.

Lemma 2.8. Let G be K_3-free with $\Delta(G) \leqslant d$, then

$$\alpha(G) \geqslant \frac{cn}{d} \log(d).$$

for some universal constant $c > 0$.

Proof. (Of the theorem assuming Lemma 2.8) Notice that G is K_3 free and its independence number is less than k, then, as every neighborhood is a independent set, we have $\Delta(G) < k$. Applying the lemma we get that

$$k > \alpha(G) \geqslant \frac{cn}{k} \log(k)$$

which implies

$$n \leqslant C \frac{k^2}{\log(k)}.$$

□

The proof of the lemma is exceptionally fun. The main idea is to take a uniform random independent set in G, and show that in expectation, its size is bigger than $\frac{cn}{d} \log(d)$ - which is very unnintuitive.

Proof. (Of Lemma 2.8) Let $I \in \mathcal{I}(G)$ be a uniformly chosen random independent set of G. For each $v \in V(G)$, define the random variable

$$X_v = d \cdot \mathbf{1}[v \in I] + \sum_{u \in N(v)} \mathbf{1}[u \in I].$$

Note that

$$\mathbb{E} \sum_{v \in V(G)} X_v = d \cdot \mathbb{E}|I| + \sum_{v \in V(G)} \sum_{u \in N(v)} \mathbb{P}(u \in I) = d \cdot \mathbb{E}|I| + \sum_{u \in V(G)} d(u) \mathbb{P}(u \in I) \leqslant 2d \cdot \mathbb{E}|I|.$$

So we have that $\mathbb{E}|I| \geqslant \frac{1}{2d} \mathbb{E}[\sum_{v \in V(G)} X_v]$. To prove the lemma, it would suffice to show then that $\mathbb{E}X_v \geqslant \log(d)$ for all v (as summing it would yield the result).

So let's lowerbound $\mathbb{E}X_v$. The idea is to condition it on something and find a bound that works **FOR EVERY** condition we take (which seems insane).

Fix v, let's divide the world into $U = \{v\} \cup N(v)$ and $W = V(G) \setminus U$ (both depend on v). Notice that the value of X_v only changes for elements in U that appear in I. So fix an independent set S of W, and let's calculate $\mathbb{E}[X_v | I \cap W = S]$.

From the fact that I has to be an independent set, we know it cannot have any neighbors of elements in S. So let $Y = N(v) \cap N(S)$ and let $X = N(v) \setminus Y$ (note $v \notin N(S)$). As I is uniformly chosen, we now know that $I \cap U$ (conditioned on $I \cap W = S$), is a uniformly

chosen independent set of $\{v\} \cup X$. Let $|X| = t$, remembering that X is an independent set, we find

$$\begin{aligned}\mathbb{E}[X_v | I \cap W = S] &= \mathbb{E}[X_v | v \in I, I \cap W = S] \cdot \mathbb{P}(v \in I | I \cap W = S) \\ &\quad + \mathbb{E}[X_v | v \notin I, I \cap W = S] \cdot \mathbb{P}(v \notin I | I \cap W = S) \\ &= \frac{1}{2^t + 1} d + \frac{2^t}{2^t + 1} \frac{t}{2}\end{aligned}$$

where we used the fact that $I \cap U$ is an uniform independent set of the star $G[\{v\} \cup X]$. Now (here comes the crazy part), notice that independent of what $t = |X|$ is, we have

$$\frac{1}{2^t + 1} d + \frac{2^t}{2^t + 1} \frac{t}{2} \geqslant \frac{\log(d)}{8}.$$

just take $t < \log(d)/2$ and $t \geqslant \log(d)/2$. But this means

$$\mathbb{E}X_v = \mathbb{E}[\mathbb{E}[X_v | I \cap W(v)]] \geqslant \frac{\log(d)}{8}$$

yielding the result. □

2.2 Bounds on R(4,k)

By doing vertex deletion and Erdös-Szekeres, we may easily find (we need $p^6 n^4 < n$, so $p = \varepsilon n^{-1/2}$)

$$c\left(\frac{k}{\log(k)}\right)^2 \leqslant R(4,k) \leqslant k^3.$$

Doing the Erdös-Tetali method [2.5] (we need $p^6 n^4 < pn^2$, so $p = \varepsilon n^{-2/5}$) we get also

$$c\left(\frac{k}{\log k}\right)^{5/2} \leqslant R(4,k).$$

But we can do better on the upperbound by doing a similar strategy as in the $R(3,k)$ case. The next lemma Rob stated, but I don't think we will use it.

Lemma 2.9. Let G on n vtx be K_4-free and $\Delta(G) \leqslant d$, then

$$\alpha(G) \geqslant \frac{cn}{d} \frac{\log(d)}{\log\log(d)}.$$

Proof. The proof of this is similar to lemma 2.8, we must however pick a uniformly independent set from the now K_3-free X. (I will try to do the computations more carefully later) □

To get an even better bound, we will need some kind of stability setting in these super sparse graphs.

Lemma 2.10. (AKSz) Let G be a graph on n-vtx with $\Delta(G) \leqslant d$ and $k_3(G) \leqslant (nd^2)/\lambda$, then

$$\alpha(G) \geqslant c\frac{n}{d}\log(\lambda)$$

where $c > 0$ is an absolute constant.

With it we can prove the following strong bound,

Theorem 2.11.

$$R(4,k) \leqslant C\frac{k^3}{\log(k)^2}.$$

Proof. (Assuming [2.10]) Let G be n-vtx, K_4-free graph with $\alpha(G) < k$, we know by Erdös-Szekeres that $\Delta(G) < R(3,k)$. How many triangles can G have? Pick a vertex v and let $S = N(v)$. We know that for any vertex $u \in S$, $N_{G[S]}(u)$ is independent (because G is K_4-free) and so $d_{G[S]}(u) < k$. This implies that

$$k_3(G) = \frac{1}{3}\sum_{v\in V(G)} e(G[N(v)]) \leqslant \frac{nk^2}{6}$$

by our bound on $\Delta(G[S])$.

So applying [2.10] with $d = R(3,k)$ and $\lambda = (k/\log(k))^2$ yields

$$k > \alpha(G) > c\frac{n\log(k)^2}{k^2}$$

which yields $n < Ck^3/\log(k)^2$. □

So now it suffices to prove 2.10.

Proof. (Of the AKSz lemma) The idea will be to do some vertex deletion and apply 2.8, but we don't really know how so we do it randomly.

To easy up the calculation (it only changes the constant in front), suppose $\Delta(G) \leqslant d$ and $k_3(G) \leqslant nd^2/\lambda^3$. Take a q-random subset Q of $V(G)$, we can estimate some statistics, we find:

(i) $\mathbb{E}|Q| = qn$

(ii) $\mathbb{E}e(G[Q]) = q^2e(G) \leqslant q^2nd/2$

(iii) $\mathbb{E}k_3(G[Q]) = q^3k_3(G) \leqslant (q/\lambda)^3nd^2$

By taking $q = \lambda/d$ and using Markov to remove triangles and high degree vtxs, we find a set of size at least $qn/6$ with no triangles and $\Delta(G[Q']) < 3q\Delta(G)$. Applying 2.8 to this set yields the result.

(Formally) We have very good concentration on $|Q|$ (as long as $qn \to \infty$), so applying Chernoof and Markov (twice) we see that

$$\begin{aligned}\mathbb{P}\left(\left(|Q| > \frac{qn}{2}\right) \wedge \left(e(G[Q]) < 3q^2 e(G)\right) \wedge \left(k_3(G[Q]) < \frac{3q^3nd^2}{\lambda^3}\right)\right) \\ = 1 - \mathbb{P}\left(\left(|Q| < \frac{qn}{2}\right) \vee \left(e(G[Q]) > 3q^2 e(G)\right) \vee \left(k_3(G[Q]) > \frac{3q^3nd^2}{\lambda^3}\right)\right) \\ \geqslant 1 - o(1) - 1/3 - 1/3 > 0\end{aligned}$$

So we find a $G' = G[Q]$ for some good Q satisfying all these properties. G' satisfies:

(i) $v(G') \geqslant \frac{qn}{2}$

(ii) $e(G') \leqslant 3q^2 e(G) \leqslant 3q^2 nd/2$

(iii) $k_3(G') \leqslant \frac{3q^3nd^2}{\lambda^3}$

By removing at most half of the vertices of G' (using Markov) and killing a vertex from each triangle, we find triangle free G'' with $v(G') \geqslant \frac{qn}{4} - k_3(G') \geqslant \frac{qn}{8}$ and $\Delta(G'') \leqslant 6q^2nd/2$. □

This seems like a very interesting strategy for off-diagonal Ramsey. We did the case $l = 3$, $l = 4$, so a very natural conjecture would be:

Theorem 2.12.

$$R(l,k) \leqslant C\frac{k^{l-1}}{(\log k)^{l-2}}$$

Proof. We prove it by induction. The case $l = 3, 4$ were already done. Let's follow the proof of [2.11]. Suppose G is an n-vtx, K_l-free graph with $\alpha(G) < k$. We must have $\Delta(G) < R(l-1,k) \leqslant d$. How many triangles can it have? We know that if $u \in N(v)$, then $|N(u) \cap N(v)| < R(l-2,k)$. So, doing the same bound as before, we see that $k_3(G) \leqslant ndR(l-2,k)/3$. So, to use 2.10 we must have

$$\frac{ndR(l-2,k)}{3} \leqslant \frac{nd^2}{\lambda}$$

so

$$\lambda \leqslant \frac{3d}{R(l-2,k)}$$

and (by induction), it suffices that λ satisfies

$$\lambda \leqslant \frac{cd(\log k)^{l-2}}{k^{l-1}}$$

applying 2.10 with $d = \frac{k^{l-1}}{(\log k)^{l-2}} \geqslant R(l-1,k) > \Delta(G)$, we have by hypohesis,

$$\alpha(G) \geqslant \frac{n}{d}\log(\lambda) \geqslant C_l \frac{n\log(k)^{l-2}}{k^{l-2}}$$

and therefore,

$$n \leqslant C_l \frac{k^{l-1}}{(\log k)^{l-2}}$$

□

This proof can also be found in Rob's ICM manuscript. [https://arxiv.org/pdf/2601.05221].

2.3 Bounds on R(3,3,k)

So what are some easy bounds we can get on this? An easy lowerbound is $R(3,k)$, by doing Erdös-Szekeres, we can get the multinomial bound as well

$$R(3,k) \leqslant R(3,3,k) \leqslant 2kR(3,k) \leqslant C\frac{k^3}{\log k}.$$

We can do a bit better on the upperbound by counting triangles and applying [2.10]!

Theorem 2.13.

$$R(3,3,k) \leqslant \frac{k^3}{(\log k)^2}$$

Proof. Suppose there is a coloring χ on K_n that shows that $R(3,3,k) \geqslant n$. Let $G = E_R(\chi(K_n)) \cup E_B(\chi(K_n))$ that is, G is the graph of red and blue edges. Note that $\alpha(G) < k$, as the coloring had no green K_k. To apply [2.10], we must calculate $k_3(G)$ and we know it does not have any full red, or any full blue triangles. So let's bound the number of $\{R,R,B\}$ triangles in G (the number of $\{B,B,R\}$ by simmetry will be the same). One way to count $\{R,R,B\}$ triangles is by picking the cherry vertex.

$$k_{\{R,R,B\}}(G) = \sum_{v\in G} e_B(G[N_R(v)])$$

By Erdös-Szekeres, we know that $N_R(v)$ is red-edge-free and $|N_R(v)| \leqslant R(3,k)$ for any vertex v. We also have that for any $u \in N_R(v)$, $N_B(u) \cap N_R(v)$ is a independent set, by the previous observation and the fact that we are blue triangle free. So $d_B[G[N_R(v)]](u) < k$ (here this

monstrosity means that the blue degree of u in $N_R(v)$ is less than k). This implies that $e_B(G[N_R(v)]) < R(3,k)k/2$ for any v and therefore that

$$k_{\{R,R,B\}}(G) < \frac{nR(3,k)k}{2}$$

so the total number of triangles is also of order $nk^3/\log k$. Applying [2.10] with $d = \frac{Ck^2}{\log k} \geqslant R(3,k)$ and this $k_3(G) < ndk$ we get that we can use $\lambda = Cd/k \geqslant \frac{Ck}{\log k}$. So we get

$$k > \alpha(G) \geqslant \frac{n}{d}\log k \geqslant \frac{C'n(\log k)^2}{k^2}$$

which implies $n \leqslant Ck^3/(\log k)^2$. □

Next class we will see a very cool lowerbound by Alon, who observed the following lemma:

Lemma 2.14. If there exists K_3-free G on n vtx with number of independent k sets of G less than $\binom{n}{k}^{1/2}$ then $R(3,3,k) > n$.

Proof. Embed G randomly in K_n with color red, do it again in color blue. By the union bound, no independent k set survive (so we win by painting non edges with green). □

We will turn for a pseudo-random construction to prove the existence of a graph that satisfies [2.14]. For this, we need a good definiton of what pseudo-random means.

Definition 2.15. We say that a n-vtx graph G is (p,β)-jumbled if $e(G) = p\binom{n}{2}$ and for every $S, T \subset V(G)$, we have that

$$|e(S,T) - p|S||T|| \leqslant \beta\sqrt{|S||T|}.$$

where here $e(S,T)$ is the number of edges that start in S and end in T. Edges in $S \cap T$ are counted twice.

The most pseudo-random graph (the best β you can get) is with $G(n,p)$, I don't know anyway to prove this now, but let's calculate how jumbled $G(n,p)$ is.

Theorem 2.16. $G(n,p)$ is (p',β)-jumbled with $p' = e(G(n,p))/\binom{n}{2} = (1+o(1))p$ and $\beta = O(\sqrt{pn})$ w.h.p.

Proof. We use Chernoff inequality [A.5], to note that for absolute $C, c > 0$ and taking A suffciently big,

$$\mathbb{P}(|e(S,T) - p|S||T|| \geqslant A\sqrt{pn|S||T|}) \leqslant C\exp(-cA^2n) \ll 4^{-n},$$

so $G(n,p)$ is $\beta = A\sqrt{np}$-jumbled w.h.p by the union bound. □

This yields the following definition

Definition 2.17. A graph is said to be **optimally pseudorandom** if it is (p, β)-jumbled with $\beta = O(\sqrt{pn})$.

The following question is at the core of this method. How many independent k-sets can a (p, β)-jumbled graph have? We now have a deterministic fact about our graph and we can try to count. I won't do it here, but the idea is to count sequences $(v_1, \ldots, v_k)$ of vertices that can make a independent set.

Using the (p, β) jumbled condition, we can count how many "bad" vertices there are when trying to grow the sequence. We can make only a few "good" steps (extension steps where our non-neighborhood decreases with correct size), because they reduce the non-neighborhood by a $(1-p)/2$ ratio which is just too fast. And each "bad" step (steps which don't reduce the non-neighborhood much) we are obliged to choose from a small set. So counting everything yields a result.

The following lemma will be stated without proof, it is an algebraic construction.

Lemma 2.18. (Quadrangle) There exists a n-vertex, $n^{1/3}$-regular, $n^{1/3}$-uniform hypergraph H that satisfies:

(i) $\forall e_1, e_2 \in E(H), \quad |e_1 \cap e_2| \neq 2.$

(ii) $\forall e_1, e_2, e_3 \in E(H)$ it does not hold that $|e_1 \cap e_2| = |e_2 \cap e_3| = |e_1 \cap e_3| = 1$.

Putting a randomly embedded $K_{t,t}$ with $t = n^{1/3}/2$ on each edge we get a K_3-free, optimally pseudorandom graph where p is the maximum it can be, $p = n^{-1/3}$. (As Rob pointed out an optimally pseudorandom, triangle-free graph must have density at most $p = n^{-1/3}$).

2.4 Ramsey of bounded degree graphs

The motivation of this section will be the following theorem.

Theorem 2.19. Let H be a bipartite graph with k vertices and $\Delta(H) \leqslant d$. Then $r(H) \leqslant 2^{d+6}k$.

2.4.1 Old Friends

To uncover it, we will need to remember some old forgotten friends. The LLL and the DRC.

Theorem 2.20 (Lovász Local Lemma)**.** Let $\mathcal{E} = \{B_1, B_2, \ldots, B_m\}$ be bad events in some probability space and suppose that for each event B_i, there exists a set $N(B_i) \subset \mathcal{E}$ such that

B_i is mutually independent of all events in $\mathcal{E} \setminus N(B_i)$. We say that $j \sim i$ if $B_j \in N(B_i)$. If there exists $x_1, \dots x_m \in (0,1)$ satisfying

$$\mathbb{P}(B_i) \leqslant x_i \prod_{j \sim i} (1 - x_j)$$

then

$$\mathbb{P}\left(\bigcap_{i=1}^{m} B_i^c \right) > \prod_{i=1}^{m} (1 - x_i).$$

Remark 2.21. Usually people talk about the dependence graph, in this framework the neighbors of an event B_i are precisely $N(B_i)$. Note that the choice of $N(B_i)$ (or of the dependece graph) is not necessarily unique.

Proof of LLL 2.20. The proof is just a very smart induction (similar to Harris or Janson - which I should also add in the appendix). We lemma that given any $S \subset [m]$,

$$\mathbb{P}\left(B_i \mid \bigcap_{j \in S} B_j^c \right) \leqslant x_i.$$

Note that this implies the theorem as we can always write

$$\mathbb{P}\left(\bigcap_{i=1}^{m} B_i^c \right) = \prod_{i=1}^{m} \mathbb{P}\left(B_i^c \mid \bigcap_{j<i} B_j^c \right)$$

and the result follows from the lemma.

Let's prove the lemma by induction on $|S|$. If $S = \varnothing$, it trivially follows as $\mathbb{P}(B_i) \leqslant x_i \prod_{j \sim i}(1 - x_j) < x_i$. Now let's separate the event

$$\bigcap_{j \in S} B_j^c = \bigcap_{\substack{j \in S \\ j \sim i}} B_j^c \cap \bigcap_{\substack{j \in S \\ j \not\sim i}} B_j^c = E \cap F$$

we think of E as the influence of the "neighbors" in the dependence graph and F as the influcence of the non-neighbors. We can now write

$$\mathbb{P}(B_i \mid E \cap F) = \frac{\mathbb{P}(B_i \cap E \cap F)}{\mathbb{P}(E \cap F)} = \frac{\mathbb{P}(E \mid B_i \cap F)\mathbb{P}(B_i \mid F)\mathbb{P}(F)}{\mathbb{P}(E \mid F)\mathbb{P}(F)} \leqslant \frac{x_i \prod_{j \sim i}(1 - x_j)}{\mathbb{P}(E \mid F)} \tag{1}$$

If we bound $\mathbb{P}(E \mid F) \leqslant 1$ and use the independence to say $\mathbb{P}(B_i \mid F)$ is less than what given by the hypothesis. Now what is $\mathbb{P}(E \mid F)$? Opening it up we get (we abuse the notation and say that $j \in E$ if E is taking the intersection with B_j^c)

$$\mathbb{P}(E \mid F) = \prod_{j \in E} \mathbb{P}\left(B_j^c \mid \bigcap_{\substack{k<j \\ k \in E}} B_k^c \cap F \right) \geqslant \prod_{j \in E} (1 - x_j)$$

by the induction hypothesis. Substituting in the inequality [1] we prove the lemma. □

Remark 2.22. We cannot take the graph to be degenerate in the symmetric case as we need the I.H. to compute $\mathbb{P}(E \mid F)$.

Remark 2.23. In the hypothesis, we didn't use full mutual independence, only that $\mathbb{P}(B_i \mid F) \leqslant \mathbb{P}(B_i)$. Using only this requirement, we call it *Lopsided LLL.*

Corollary 2.24 (Symmetric LLL)**.** Suppose that in the hypothesis of [2.20], we had $\max_i |N(B_i)| \leqslant d$. That is the dependence graph D is such that $\Delta(D) \leqslant d$. Suppose moreover that for all i,

$$\mathbb{P}(B_i) \leqslant \frac{1}{e(d+1)}$$

then $\mathbb{P}(\bigcap_{i \leqslant m} B_i^c) > 0$.

Proof. Take $x_i = 1/(d+1)$, and note that

$$\mathbb{P}(B_i) \leqslant \frac{1}{e(d+1)} \leqslant \frac{1}{(d+1)}\left(1 - \frac{1}{d+1}\right)^d \leqslant x_i \prod_{j \sim i}(1 - x_j).$$

□

The LLL is a pearl that should always be within our hearts. But there is a even simpler theorem which we must never forsake.

Theorem 2.25 (Dependent Random Choice)**.** Let $k, s \in \mathbb{N}$ and G be any graph with n vtxs. For any $t \in \mathbb{N}$, there exists a set $U \subset V(G)$ with size greater or equal to

$$\frac{(2e(G))^t}{n^{2t-1}} - \binom{n}{k}\left(\frac{s}{n}\right)^t$$

such that every set of k vertices of U has s neighbors in common.

Proof. Choose independently and uniformly t vertices from $V(G)$ with possible repetition. Call this random set T and let $X = |N(T)|$. Let Y be the number of k-sets in $N(T)$ with less than s common neighbors. Let's compute $\mathbb{E}X$ and $\mathbb{E}Y$. We may write

$$X = \sum_{v \in V(G)} \mathbf{1}[v \in N(T)]$$

So $\mathbb{E}X = \sum_{v \in V(G)} \mathbb{P}(v \in N(T))$. Now if $v \in N(T)$, then every vertex in T belongs to $N(v)$, as we chose u.i.i.d we find

$$\sum_{v \in V(G)} \mathbb{P}(v \in N(T)) = \sum_{v \in V(G)} \mathbb{P}(T \subset N(v)) = \sum_{v \in V(G)} \left(\frac{d(v)}{n}\right)^t \geqslant n\left(\frac{2e(G)}{n^2}\right)^t$$

where the last inequality follows by Jensen.

Now what about Y? To ease notation, let (only this once) $(V(G))_{[k]} = \binom{V(G)}{k}$, then

$$Y = \sum_{\substack{S \in (V(G))_{[k]} \\ |N(S)|<s}} \mathbf{1}[S \subset N(T)] = \sum_{\substack{S \in (V(G))_{[k]} \\ |N(S)|<s}} \mathbf{1}[T \subset N(S)]$$

taking expectation and upperbounding by all k-sets, we get

$$E[Y] \leqslant \binom{n}{k}\left(\frac{s}{n}\right)^t.$$

We have that

$$\mathbb{E}[X - Y] \geqslant \frac{(2e(G))^t}{n^{2t-1}} - \binom{n}{k}\left(\frac{s}{n}\right)^t = a \tag{2}$$

so there exists an instance of T, such that by removing a vertex from each bad k-set in $N(T)$, we are still left with a vertices. □

2.4.2 Applications

Let's use the LLL [2.20] to prove lowerbounds on Ramsey.

Erdös showed that, in his majestic use of the probabilistic method, the following theorem.

Theorem 2.26. Consider a random coloring of the edges of K_N in red and blue, each chosen independently and uniformly. If X is the number of monochromatic cliques of size k and $\mathbb{E}X < 1$, then $R(k) > N$. In particular, $R(k) \gtrsim 2^{k/2}$.

Proof. The first statement is a proof in itself, maybe the most beautiful proof of all. For the second we only need to do the computation. We know

$$\mathbb{E}X = \binom{n}{k} 2^{-\binom{k}{2}+1} \leqslant \left(\frac{en}{k} 2^{-k/2}\right)^k.$$

(We lose a bit when doing the binom approximation and we use it). By choosing n appropriately, this yields

$$R(k) > \frac{k}{e} 2^{k/2}.$$

□

Spencer got a crazy improvement, unbeaten by the ages (50 years already), using the LLL.

Theorem 2.27. (Spencer 77)

$$R(k) > \frac{2k}{e} 2^{k/2}.$$

He also managed to prove Erdös lowerbound on $R(3, k)$ in a much simpler way - way before Krivilevich's proof [see Theorem 2.4].

2.4.3 Proof of Conlon-Fox-Sudakov

We will now try to prove Theorem [2.19] using a twisted combination of our old friends. It's a very nice proof.

The first method that comes to mind when trying to find bipartite graphs is DRC, but it alone won't be strong enough to give us the upperbound we want. So let's do something a bit better.

Lemma 2.28. Let G be a graph on n vtxs and suppose $e(G) \geqslant c\binom{n}{2}$ for some $c \leqslant 1/2$. Suppose moreover that $d, k \in \mathbb{N}$ are such that

$$2d \leqslant k \leqslant c^{d+2}n.$$

Then, there exists a set $A \subset V(G)$ satisfying

(a) $|A| \geqslant k$

(b) At most $c^d\binom{|A|}{d}$ d-subsets of A have less than k neighbors in common.

Proof. The proof of this will be exactly the same as the proof of DRC [2.25] but we will take expectation over a different expression [2]. Let $T = (u_1, \dots u_t)$ be a random uniformly independently chosen set of vertices. As before we set $X = |N(T)|$, and Y the number of bad d-sets. Again we have that

$$\mathbb{E}X \geqslant c^t n \quad \text{and} \quad \mathbb{E}Y \leqslant \binom{n}{d}\left(\frac{k}{n}\right)^t$$

Crucially, we want to show that with positive probability $X \geqslant k$ and $Y \leqslant c^d\binom{|X|}{d}$. So we will take the mean of a magical r.v with positive expectation. Consider Z given by

$$Z = X^t - \frac{(\mathbb{E}X)^t}{2} - \frac{(\mathbb{E}X)^t}{2}\frac{Y}{\mathbb{E}Y}.$$

Note that as $X, Y \geqslant 0$, we know that $\mathbb{E}X^t \geqslant (\mathbb{E}X)^t$, so

$$\mathbb{E}Z = \mathbb{E}X^t - \frac{\mathbb{E}[X]^t}{2} - \frac{\mathbb{E}[X]^t}{2} \geqslant 0.$$

As both $(\mathbb{E}X)^t/2$ and $(\mathbb{E}X)^t Y/(2\mathbb{E}Y)$ are positive, it follows that with positive probability $X^t \geqslant (\mathbb{E}X)^t/2$ and, most importantly,

$$X^t \geqslant \frac{(\mathbb{E}X)^t}{2}\frac{Y}{\mathbb{E}Y}.$$

So we found that with probability > 0,

$$Y \leqslant \binom{n}{d}\left(\frac{k}{n}\right)^t \frac{2X^t}{c^{t^2}n^t} \leqslant \binom{n}{d}\frac{2c^{t(d+2)}X^t}{c^{t^2}n^t}$$

taking $t = d$, we get

$$Y \leqslant \frac{c^{2d}X^d}{d!} \leqslant c^d \binom{X}{d}$$

if d is big enough. The other bound follows easily,

$$X \geqslant \frac{\mathbb{E}X}{2^{1/d}} \geqslant \frac{c^d n}{2^{1/d}} \geqslant k.$$

□

With this new lemma, let's prove a weaker version of [2.19] (maybe I can see later how the stronger one follows.)

Theorem 2.29. Let H be a d-regular bipartite graph on k vtxs. Then

$$r(H) \leqslant 2^{d+5}k.$$

Proof. Let $n \geqslant 2^{d+5}k$ and G a graph on n vtxs with $e(G) \geqslant \frac{1}{2}\binom{n}{2}$ - this is a density proof, actually we find H in the densest color. Suppose $H = A \sqcup B$ and apply lemma [2.28] to G with $c = 1/2$. We find a set A^* satisfying the conclusions of the lemma. We will use the LLL [2.20] to randomly embed A into A^* in such a way that for every $b \in B$, $N(b)$ is sent to a good set in A^*. Because of this, we can trivially find a copy of H in G.

To really use the LLL, we cannot actually embed it ramdomly, this would kill all the mutual independence of the events. What we actually do is a random assignment.

Let $f : A \to A^*$ be a random map. We will define two types of bad events, collisions and bad sets. For each $u, v \in A$, define the events

$$F_{u,v} = \{f(u) = f(v)\}$$

and for each b in B, define

$$E_b = \{f(N(b)) \text{ is a bad } d\text{-set}\}.$$

Amazingly here, if non of the $F_{u,v}$ occur, no neighborhood is mapped to less than d elements, so we don't even need to care that $f(N(b))$ has the right size. Now for the dependence graph, by how we chose our probability space, we have that disjoint events (on the vertex of H) are non-edges:

- $F_{u,v} \sim F_{u',v'}$ if $\{u, v\} \cap \{u', v'\} \neq \varnothing$.
- $F_{u,v} \sim E_b$ if $\{u, v\} \cap N(b) \neq \varnothing$.
- $E_b \sim E_{b'}$ if $N(b) \cap N(b') \neq \varnothing$.

We need to find x_F and x_E satisfying the hypothesis of 2.20. We know that

$$\mathbb{P}(F_{u,v}) = \frac{1}{|A^*|} \leqslant \frac{1}{k}$$

and

$$\mathbb{P}(E_b) \leqslant c^d.$$

By counting the neighborhoods of the events, we need to satisfy

$$\mathbb{P}(E_b) \leqslant c^d \leqslant x_E(1-x_E)^{d^2}(1-x_F)^{d|A|}$$
$$\mathbb{P}(F_{u,v}) \leqslant \frac{1}{|A^*|} \leqslant x_F(1-x_E)^{2d}(1-x_F)^{2|A|}$$

so we really should check which is biting us by trying both x_E close to $\mathbb{P}(E_b)$ or x_F close to $\mathbb{P}(F_{u,v})$. In this case, the collisions are biting. Let $x_F = 2/|A^*|$, then (we might have to take A^* bigger here this is why we get a 2^{d+6}, we do $|A^*| = 8|A|$)

$$\frac{1}{|A^*|} \leqslant \frac{2}{|A^*|}(1-x_E)^{2d}e^{-2|A|/|A^*|},$$

So we need

$$(1-x_E)^{2d} \geqslant \frac{1}{2e^{1/8}}.$$

From the first inequality we also get

$$c^d \leqslant x_E(1-x_E)^{d^2}e^{-d|A|/|A|^*} \leqslant x_E e^{-x_E d^2}e^{-d/8}$$

so taking $x_E = (ce^{1/8})^d$ we get the result when d tends to infinity.

□

2.5 Erdős-Rogers numbers and Open Problems

Let's look into the very nice generalization of Ramsey numbers, the Erdős-Rogers numbers $f_{H,G}(n)$. Prepare for the mouthful definition.

Definition 2.30. Given graphs H and F, we let $f_{H,F}(n)$ be the maximum k such that every F-free graph G on n vertices contains a H-free set of size k. In symbolic terms

$$f_{H,F}(n) = \min_{\substack{v(G)=n \\ F\not\subset G}} \max_{\substack{U\subset V(G) \\ H\not\subset G[U]}} |U|$$

So we are interested in graphs that are F-free but have the highest density (and spread out too) of copies of H. For this reason, the function was most studied for $H = K_s$ and $F = K_{s+1}$. And this case, which we usually denote by $f_{s,s+1}(n)$ was solved assymptoptically by Rob and Julian!

Theorem 2.31.

$$f_{s,s+1}(n) = \Theta(\sqrt{n \log n})$$

The lowerbound was proved using a smart theorem on the clique chromatic number of a graph. The upperbound (proved by Rob and Julian) showed up after using a smart (Hefty-Horn-King-Pfender) construction.

Some numbers are already known. From the recent improvements in off-diagonal ramsey, we get that

$$f_{K_2,K_s}(n) \lesssim n^{1/(s-1)+o(1)}$$

(this is actually equivalent to the ramsey numbers so this is basically tight). Although we have this result, much much less is known about f_{K_3,K_s}. Using DRC [2.25] Benny Sudakov, managed to show

Theorem 2.32.

$$f_{K_3,K_5} = \Omega(n^{5/12}).$$

Note that, in order to prove something like this. That $f_{K_3,K_5}(n) \geqslant k(n)$, we must show that in every graph G with the property that every k set has a triangle, we can find a K_5 (this is not really obvious but follows from the definition). Let's quickly prove this theorem.

We start off by saying that if our graph has few triangles, then we can easily find a big set without triangles and contradict the hypothesis. The following lemma follows from a simple first moment bound.

Lemma 2.33. Let G be a graph on n vertex and dn edges. Suppose further that $k_3(G) = m$ and p is such that

$$p^3 m \leqslant pn$$

then we can find a K_3 set of size $pn - p^3 m$.

As G is K_5-free by doing erdos-szekeres, we can bound the number of K_3's to be at most ndk. Using the previous lemma (and setting $p = 1/\sqrt{dk}$) we see that we must have $dn > (n/k)^3$. So now that we have a lowerbound on the number of edges of our graph, we can apply DRC to check that if there were $t \in \mathbb{N}$ satisfying

$$n\left(\frac{d}{n}\right)^t - \binom{n}{2}\left(\frac{k}{n}\right)^t > k$$

we could find a set of size k such that every pair of elements contain a shared neighborhood of size k as well. Becase by assumption every set of size k contains a triangle, we could easily find our K_5. Solving this equation for $t = 1$ or $t = 2$ and putting $d/n = n/k^3$, we get that if $k < n^{-5/12}$ or else we could find a K_5.

Gowers and O. Janzer showed doing basically the blow up construction with some smart deletion that $f_{3,5}(n) = O(n^{6/13})$. Very recently Janzer and Sudakov showed using the magical

Hermitian Unital and putting blow ups of $K_{t,t,\dots,t}$ (s times) in their edges that $f_(s, s+2)(n) = O(n^{\frac{2s-3}{4s-5}})$ I should even read this, seems cool!

Not much more is known! It's the open wild west in here!

2.6 Induced Ramsey Numbers

In this section we will prove some results in induced ramsey numbers. It is motivated by the amazing breakthrough of Marcelo, Lucas, Rafael e João who showed (with a very technical proof) that $R_{\text{ind}}(H) \leqslant 2^{Cv(H)}$ for some universal C and every graph H.

I won't be able to give even a faint resemblance of their proof, so let's focus on easier results that came before. First of all, we should define our object.

Given a finite graph H, we call $R_{\text{ind}}(H)$ as the minimum N, such that there exists a graph G on n vertices s.t every 2-coloring of the edges of G contains an induced monochromatic copy of H. It is a mouthful!

It is not clear even that this is finite. What is most amazing is that $G(N, 1/2)$ works for every H if N is big enough!

Theorem 2.34. Let H be a graph on k vertices and $N \geqslant N_0(k) = O(k^2)$ be sufficiently big. Then, with positive probability, every 2-coloring of the edges $G(N, 1/2)$ contains an induced monochromatic copy of H.

The idea here is to use an assymetric container lemma for colorings! More precisely, we want to show

$$\mathbb{P}(\exists c : E(G(N,1/2)) \to \{R, B\} \text{ without induced monochromatic copy of } H) \leqslant \sum_{C \in \mathcal{C}} \mathbb{P}(...) \to 0$$

To deal with induced copies we will define a smart hypergraph that encodes them. Let $\mathcal{H}$ ve a bipartite hypergraph with vertex parts V_0 and V_1 each being a copy of $E(K_n)$. We will think of V_0 as encoding the non-edges of our graph and V_1 encoding the edges. Let $E(\mathcal{H})$ be the sets that encode an induced copy of H. That is $e \in E(\mathcal{H})$ if $e \cap V_1$ are the edges of some copy of H and $e \cap V_0$ are exactly the non-edges. Note that if $G \subset K_n$ contains no induced copy of H, then (G^c, G) is an independent set in $\mathcal{H}$.

Associated with this hypergraph, Balogh and Samotij proved the following container theorem

Theorem 2.35. For every H and its associated $\mathcal{H}$, there exists a family of containers $\mathcal{C}$ satisfying

1. $|\mathcal{C}| \leqslant \exp(O_k(N^2))$
2. $\forall (G_0, G_1)$ not containing and induced H, there exists (C_0, C_1) with $G_0 \subset C_0$ and $G_1 \subset C_1$.

3. (C_0, C_1) contain $o(n^k)$ induced copies of H for every choice of containers.

Now the idea is as usual, we show that if C_0 and C_1 are small a bit small, than $G(N, 1/2)$ must be avoiding a positive proportion of edges which happens with super small probability. On the other case that they are big, we will find many induced monochromatic copies, providing a contradiction.

Proof. Note that any coloring of the edges of $G(N, 1/2)$ in red and blue yields a triple (G^c, G_R, G_B) of disjoint edges. If our coloring is free of monochromatic induced H, it must be that (G^c, G_R) and (G^c, G_B) are independent sets in $\mathcal{H}$. In particular

$$\mathbb{P}(\exists G_B \cup G_R = G, \text{ no induced MC } H) \leqslant \sum_{(C_0, C_B, C_R)} \mathbb{P}(G^c \subset C_0, G_B \subset C_B \cup G_R \subset C_R).$$

If $|C_0| \leqslant (1-\delta)\binom{n}{2}$ or $|C_B \cup C_R| \leqslant (1-\delta)\binom{n}{2}$ we are very happy, as it would imply $G(N, 1/2)$ avoids a positive proportion of edges.

Let's show that this is always the case. Suppose for sake of contradiction that $|C_0|$ and $|C_B \cup C_R|$ are big. In particular thinking of these containers as sets of $E(K_n)$, we have that $|C_0 \cap (C_B \cup C_R)| \geqslant (1 - 2\delta)\binom{n}{2}$. Let $k = v(H)$, by supersaturation we have $\Omega(n^{R(k)})$ copies of $K_{R(k)}$ with these edges. Because they have color red or blue, each of them contain a MC copy of K_k, yielding $\Omega(n^k)$ copies of those. Now, all these MC copies also belong to C_0, so we can choose the non-edges in such way that they make up induced copies. This contradicts our choice of containers! After all, we get

$$\mathbb{P}(\exists G_B \cup G_R = G, \text{ no induced MC } H) \leqslant \exp(o_k(N^2))2^{-\delta N^2} \to 0.$$

finishing the proof! □

3 The Method of Containers

3.1 Some nice problems

There are many reasons to understand the method of hypergraph containers. Here are some cool problems we may be interested in.

Problem 3.1. (Extremal questions in $G(n, p)$). We define

$$\mathrm{ex}(G, H) := \max\{e(G') : G' \subset G, \ G \text{ is } H - \text{free}\}.$$

Note that $\mathrm{ex}(n, H) = \mathrm{ex}(K_n, H)$. What can we say about $\mathrm{ex}(G(n, p), K_3)$?

Some easy bounds can be found. Given any sampling of $G(n,p)$, we can always find a bipartite graph with half the edges. Similarly another strategy is just to kill off triangles by removing an edge from each. This already gives

$$\mathrm{ex}(G(n,p),K_3) \geqslant \begin{cases} e(G(n,p))/2 \\ e(G(n,p)) - k_3(G(n,p)) \end{cases}$$

by finding the best regime for p in each we get

$$\mathrm{ex}(G(n,p),K_3) \geqslant \begin{cases} pn^2/4 & \text{for any } p \in [0,1] \\ (1-o(1))pn^2/2 & \text{for } p \ll \frac{1}{\sqrt{n}} \end{cases}$$

it turns out this is sharp.

Problem 3.2. When does $G(n,p) \to K_3$? Define

$$p^*(n) = \inf_{p\in[0,1]} \{p : \mathbb{P}(G(n,p) \to K_3) \geqslant 1/2\}.$$

can we find this threshold?

We know that if $G(n,p)$ contains a K_6 we are done by Ramsey so $p^* \lesssim n^{-2/5}$. But is this it? Should avoiding K_6 be enough or is there some deeper more complex structure - the answer is obviously yes.

Problem 3.3. Let $A \subset [n]_p$ with $|A| \geqslant \delta np$, does A contain a 3-AP?

Doing deletion we get that it is true if $AP_3([n]_p) = o(np)$. We know $AP_3([n]_p) \approx n^2p^3$, so if $p \ll 1/\sqrt{n}$ we get the result. What happens if $p \gg 1/\sqrt{n}$?

Very related to 3.4 is question:

Problem 3.4. How many K_3-free graphs on n vertices with m edges are there?

If we call this quantity $X(n,m)$, we know (because we can always pick from a bipartite graph) that

$$X(n,m) \geqslant \binom{n^2/4}{m}.$$

If m is sufficiently low, we can use $G(n,p)$ for our advantage! We know that

$$\mathbb{P}(K_3 \not\subset G(n,p)) \approx \exp(-n^3p^3)$$

if $p \ll n^{-1/2}$ by Janson and Harris inequality. Noting that $G(n,m) \approx G(n,p)$ with $p = m/\binom{n}{2}$, we get something that says

$$X(n,m) \approx \exp(-m^3/n^3)\binom{\binom{n}{2}}{m} \quad \text{if } m \ll n^{3/2}.$$

Is the first bound optimal?

3.1.1 Attempts at handling them

Let's follow a naive approach for tackling 3.1. So let's see first moment. Let

$$Y = \#\big\{G : K_3 \not\subset G \subset G(n,p) \text{ with } e(G) > m = (1/2+\varepsilon)p\binom{n}{2}\big\}$$

the number of such bad subgraphs of $G(n,p)$. We would like to show $Y = 0$ with high probability if $p \gg 1/\sqrt{n}$. If we calculate $\mathbb{E}Y$ we get

$$\mathbb{E}Y = \sum_{\substack{K_3 \not\subset G \\ e(G)=m}} \mathbb{P}(G \subset G(n,p)) \geqslant \binom{n^2/4}{m} p^m \eqsim \left(\frac{en^2p}{4m}\right)^m \eqsim e^m$$

where for the lowerbound we only considered the subsets of a single bipartite graph. But $\mathbb{E}Y$ is **HUGE**, so can we better this?

A very interesting point made by Rob is to look where we are losing here. Every once and while, when you look at an equation and it explodes, it is a good idea to see why. Maybe we could handle the bad case differently.

Here we are losing a ton because there are many subsets of the bipartite graph, but this seems a bit unrealistic, a bipartite graph contains at most $1/2$ of the edges of $G(n,p)$ and here we are asking for an ε deviation. To be more precise, let's check another thing.

What can we say about bipartite H in $G(n,p)$ with m edges? Can we find many of those? Note that every bipartite graph H is a subset of a complete bipartite G which has at most $n^2/4$ edges. So we could say

$$\mathbb{P}(\exists \text{ bip. } H \subset G(n,p) \text{ with } e(H) = m) = \sum_{\substack{K_{(n/2,n/2)} \sim G \\ G \subset G(n,p)}} \mathbb{P}(\exists \text{ bip. } H \subset G \cap G(n,p) \text{ with } e(H) = m)$$

It might look that we made our lives worse, but in reality this is a way simpler thing. For such H to exists, $e(G \cap G(n,p)) \geqslant m$, so here we can already apply a deviation bound. Furthermore, the number of complete bipartites is 2^{n-1} so applying the union bound seems like a great idea. We get (using A.5)

$$\begin{aligned}\sum_{\substack{K_{(n/2,n/2)} \sim G \\ G \subset G(n,p)}} \mathbb{P}(\exists \text{ bip. } H \subset G \cap G(n,p) \text{ with } e(H) = m) &\leqslant 2^n \mathbb{P}(\mathrm{Bin}(n^2/4, p) \geqslant m) \\ &\leqslant 2^n \exp(-\varepsilon^2 n^2 p/4) \to 0\end{aligned}$$

if $p \gg 1/n$ so we win by miles in this bipartite case.

What we actually did here is a container theorem for bipartite graphs. Let $\mathcal{G}$ be the family of complete bipartite graphs on n vtxs, we know that

(a) $|\mathcal{G}|$ is small $\leqslant 2^n$.

(b) Every $G \in \mathcal{G}$ is bipartite.

(c) For every bipartite $H \subset K_n$, there exists $G \in \mathcal{G}$ such that $H \subset G$. (G is the container for H).

This family allowed us to do way more efficient union bound. And this is what will try to do to tackle 3.1.

For our problem, this we need another container theorem which we will prove later:

Theorem 3.1. (Container theorem for K_3-free graphs). There exists a family $\mathcal{G}$ of graphs on n-vtxs, s.t

(a) $\mathcal{G}$ is small, $|\mathcal{G}| \leqslant 2^{n^{3/2}(\log n)^c}$.

(b) Every $G \in \mathcal{G}$ has "few" triangles, $o(n^3)$.

(c) For every K_3-free graph H, there exists a container $G \in \mathcal{G}$ s.t. $H \subset G$.

An answer for 3.1. Note that we can upperbound $\mathbb{P}(\exists K_3 \not\subset H \subset G(n,p), e(H) = m)$ using our containers. How many edges can our containers have? Because they have $o(n^3)$ triangles, by supersaturation they must have at most $(1/2 + o(1))\binom{n}{2}$ edges. So following the same calculation as before (doing the union bound and applying our deviation result), we get

$$\mathbb{P}(\exists K_3 \not\subset H \subset G(n,p), e(H) = m) \leqslant \sum_{G \in \mathcal{G}} \mathbb{P}(e(G \cap G(n,p)) \geqslant m) \leqslant 2^{n^{3/2}\log(n)^c} e^{-\varepsilon p n^2} \to 0$$

if $p \gg \log(n)^c/\sqrt{n}$. □

Next time we will try to prove Theorem 3.1, and we will do so in greater generality. We will do it for 3-uniform hypergraphs and apply it for a "triangle encoding" hypergraph.

Definition 3.2. The triangle enconding hypergraph $\mathcal{H}_{K_3}$ is a 3-uniform hypergraph with $V(\mathcal{H}_{k_3}) = E(K_n)$ and $E(\mathcal{H}_{k_3}) = \{\{e,f,g\} : e,f,g \in E(K_n) \text{ form a triangle}\}$.

3.2 A general hypergraph container statement with fingerprints

Lemma 3.3. (Hypergraph Container Lemma) For all $c \geqslant 1$ and $k \in \mathbb{N}$, there exists $\delta \sim 1/ck^4$ s.t the following holds. Let $\mathcal{H}$ be a k-uniform hypergraph on n-vtxs and $\tau > 0$ satisfying

$$\Delta_l(\mathcal{H}) \leqslant c\tau^{l-1}\frac{e(\mathcal{H})}{v(\mathcal{H})} = c\tau^{l-1}d$$

then there exists families $\mathcal{S} \subset \mathcal{I}(\mathcal{H})$ and $\mathcal{C} \subset P(V(\mathcal{H}))$ and a function $f : \mathcal{S} \to \mathcal{C}$ such that

(a) For every $S \in \mathcal{S}$, $|S| \leqslant \tau n$.

(b) For every $C \in \mathcal{C}$, $|C| \leqslant (1-\delta)n$.

(c) For every $I \in \mathcal{I}(\mathcal{H})$, there exists $S(I) \in \mathcal{S}$ s.t

$$S(I) \subset I \subset C(I) = f(S(I)).$$

With its full generality, we can prove the container theorem for K_3 encoding graphs.

Theorem 3.4. (Containers for triangle free graphs) For every $\varepsilon > 0$, there exists a family of fingerprints $\mathcal{S}(n,\varepsilon)$ and $\mathcal{G}(n,\varepsilon)$ of subgraphs of K_n satisfying the following properties:

(a) $|\mathcal{G}| \leqslant \exp(C(\varepsilon)n^{3/2}\log n)$.

(b) For every $G \in \mathcal{G}$, $k_3(G) \leqslant \varepsilon n^3$.

(c) For every triangle free graph H on n vtx, there exists $G \in \mathcal{G}$ with $H \subset G$.

Proof. We will iteratively apply our lemma [3.3] until we are left with our desired family. What degree properties can we say about $\mathcal{H} = \mathcal{H}_{K_3}(K_n)$? It's simple to note that $e(\mathcal{H}) = \binom{n}{3}$, $d = \Delta_1(\mathcal{H}) = n-2$ and $\Delta_2(\mathcal{H}) \leqslant 1$. So, the first time, we can apply the lemma with $\tau = 1/\sqrt{n}$.

Let $\mathcal{C} = \{E(K_n)\}$, while there is an element $E(G)$ in $\mathcal{C}$ with more than εn^3 triangles, we apply the container lemma on G, and substitute G by its containers in $\mathcal{C}$. We know the maximum degree condition is the same, and because $k_3(G) \geqslant \varepsilon n^3$, we know that the avg. degree in our container lemma is at least εn. So we can always repeat this process with $c = 1$ and $\tau = 1/\varepsilon\sqrt{n}$.

How many steps can we do? Well after t iterations, each container has size at most $(1-\delta)^t\binom{n}{2}$ yielding a maximum of $(1-\delta)^t n^3$ triangles. So we must have that $(1-\delta)^t \geqslant \varepsilon$, and therefore $t \leqslant \log(1/\varepsilon)/\delta$. So how many containers do we end up with? Something like

$$\binom{n^2}{n^{3/2}}^t = n^{O(n^{3/2})} = \exp(O(n^{3/2}\log n)).$$

We can even bound the sizes of the fingerprints by $t\tau\binom{n}{2} = Cn^{3/2}$. □

3.3 Some cool applications!

3.3.1 Counting triangle-free graphs

We will try to prove the following theorem:

Theorem 3.5. Almost all triangle free graphs with n vtxs and $m \gg n^{3/2}$ edges are $o(m)$ close to being bipartite.

To prove it we will need an estability result (basically Erdös-Simonitivs) that states

Theorem 3.6. If G is t-far from being bipartite, then

$$k_3(G) \geqslant \frac{n}{6}(e(G) + t - \frac{n^2}{4})$$

Proof. (of 3.5) Let $X(n, m, \alpha)$ be the number of triangle free graphs with m edges which are αm-far from being bipartite. We will upperbound X counting by counting inside the containers. So suppose we applied [3.4] with $\varepsilon > 0$, which yielded $\mathcal{G}$. We have

$$X(n, m, \alpha) \leqslant \sum_{G \in \mathcal{G}} \#\{K_3 \not\subset H \subset G,\ e(H) = m,\ H\ \alpha m\text{-far from being bip.}\}.$$

Now fix some parameter $\delta > 7\varepsilon$ we choose later, we know from the stability result that either G is δn^2 close to being bipartite or

$$e(G) \leqslant \frac{n^2}{4} + 6\varepsilon n^2 - \delta n^2 \leqslant \frac{n^2}{4} - \varepsilon n^2.$$

So let's separate our counting depending on the container and bound the previous sum

$$X(n, m, \alpha) \leqslant \sum_{\substack{G \in \mathcal{G} \\ e(G) \leqslant (1/4 - \varepsilon)n^2}} \binom{e(G)}{m} \quad + \sum_{\substack{G \in \mathcal{G} \\ G\,\delta\text{-close to bip.}}} \binom{\delta n^2}{\alpha m}\binom{n^2/4}{(1-\alpha)m}$$

where the first inequality follows from taking all m subgraphs of G when G has few edges and the second follows from choosing the αm edges that make H far from being bipartite. A bit of manipulation shows

$$\begin{aligned} X(n, m, \alpha) &\leqslant \sum_{G \in \mathcal{G}} (1 - 4\varepsilon)^m \binom{n^2/4}{m} + \binom{\delta n^2}{\alpha m}\binom{n^2/4}{m}\left(\frac{n^2}{4}\right)^{-\alpha m} \\ &\leqslant n^{Cn^{3/2}} e^{-cm} \binom{n^2/4}{m} \to o(1) \end{aligned}$$

when $m \gg n^{3/2} \log n$.

What if we want to remove the $\log n$? Then we do the same thing but we use our fingerprints, we sum over them and remember that each S must be cointained in H. Taking that into consideration the $\log n$ vanishes.

$$X(n, m, \alpha) \leqslant \sum_{|S|=0}^{Cn^{3/2}} \binom{n^2/2}{|S|}\left(\binom{(1/4 - \varepsilon)n^2}{m - |S|} + \binom{n^2/4}{(1 - \alpha/2)m - |S|}\right)$$

and after some more careful computation we win. □

3.3.2 When does Gnp arrows a triangle?

The following is a very surprising theorem (and beautiful one as well).

Theorem 3.7. Let fix $r \in \mathbb{N}$ and let $p \gg n^{-1/2}$, then $G(n,p) \to_r K_3$ w.h.p.

As always, we need some kind of saturation result. We will actually show that $\mathbb{P}(G(n,p) \not\to K_3) \to 0$. So what does this event really mean? If $G(n,p) \not\to_r K_3$, then there exists a partition (or coloring) of the edges of $G(n,p) = H_1 \cup H_2 \cdots \cup H_r$, such that each H_i is K_3-free. Now an easy double counting yields the following lemma:

Lemma 3.8. Let G be a graph and χ an r-coloring of its edges. Then if

$$e(G) \geqslant (1-\delta)\binom{n}{2}$$

then χ contains δn^3 monochromatic triangles.

And we are ready to apply our containers.

Proof. (of 3.7) We upperbound again on our containers, let C_i be the containers for H_i. We know $E(K_n) = F \cup C_1 \cdots \cup C_r$ where $k_3(C_i) \leqslant \varepsilon n^3$ for each i and F is the complement of the union. By the previous lemma, it must be that $e(F) \geqslant \varepsilon r\binom{n}{2}$, and this means that, if $G(n,p) \subset C_1 \cup \cdots \cup C_r$, it is avoiding a positive density set of edges. We find that (maximizing by the last term)

$$\begin{aligned}
\mathbb{P}(G(n,p) \not\to K_3) &\leqslant \sum_{S_1 \cup \cdots \cup S_r} \mathbb{P}((S_1 \cup \ldots S_r) \subset G(n,p) \subset (C_1 \cup \cdots \cup C_r)) \\
&\leqslant \sum_{t=0}^{Crn^{3/2}} \binom{n^2/2}{t} p^t (1-p)^{\delta n^2} \leqslant e^{-p\delta n^2} \left(\frac{epn^2}{Crn^{3/2}}\right)^{Crn^{3/2}} \\
&= \left(\frac{e^{-\delta p\sqrt{n}} p\sqrt{n}}{D}\right)^{Crn^{3/2}}
\end{aligned}$$

and it is clear that if $p\sqrt{n} \to \infty$, then the probability is tends to 0. □

3.4 A modern approach to Containers

We've seen some simple applications of the Hypergraph Container Lemma [3.3] but no proof of it. In this section, our objective will be to develop a new and modern proof (beyond the Kleitman-Winston algorithm).

3.4.1 Graph Containers

The following theorem is similar to what you get when you apply [3.3] for graphs.

Theorem 3.9 (Graph Containers)**.** Let $0 < p < \alpha < 1$, given a graph G with vtxs set V and $|V| = n$, there exists a family of containers $\mathcal{C} \subset 2^V$, satisfying the following properties:

(i) $|\mathcal{C}| \leqslant \binom{n}{pn/\alpha}$

(ii) for all $I \in \mathcal{I}(G)$, there exists $C \in \mathcal{C}$ s.t $I \subset C$

(iii) for every $C \in \mathcal{C}$, $e(G[C]) \leqslant \frac{4\alpha|C|}{p}$

We know the usual (naive) way to prove this.

Sketch of proof. For any independent set I, we start off with an empty fingerprint set $S = \varnothing$ and the whole vtx set V as C. We iteratively query each vertex in the highest remaining degree order. If it belongs to I, we add it to S and remove its neighbors from C, if not we remove it from C. A simple counting then yields properties we want if we stop S at size at most pn/α. □

Marcelos's approach will be very similar to this (although considerably more complicated), we will still query the vtxs of the graph in a smart order, but using the best information to do so! We add vertices that most reduce the space of independent sets they belong to. In some sense, the best possible fingerprints you could hope for if you chose randomly.

Instead of the maximal degree in the remaining world, we will consider the following magic statistic of a vertex v

$$\mathbb{P}(v \in C_p \,|\, C_p \in \mathcal{I}(G[C]))$$

where C_p is a p-random subset of C, which states roughly in how many independent sets the vertex v belongs to in $G[C]$. If it belongs to few, it is a good fingerprint candidate. Note that by Harris, as one event is increasing, and the other decreasing,

$$\mathbb{P}(v \in C_p \,|\, C_p \in \mathcal{I}(G[C])) = \frac{\mathbb{P}(v \in C_p \wedge C_p \in \mathcal{I}(G[C]))}{\mathbb{P}(C_p \in \mathcal{I}(G[C]))} \leqslant \mathbb{P}(v \in C_p) = p.$$

This motivates the following algorithm: Given $I \in \mathcal{I}(G)$:

1. Initialize $S = \varnothing, C = V$.
2. While $\exists v \in C$ s.t $\mathbb{P}(v \in C_p \,|\, C_p \in \mathcal{I}(G[C])) \leqslant (1-\alpha)p$ (pick v minimal)
   (a) If $v \in I$: $S \leftarrow S \cup \{v\}$, $C \leftarrow C \setminus N(v)$.
   (b) If $v \notin I$: $C \leftarrow C \setminus \{v\}$.

We need to check that in the end, this algorithm yields what we want: a deterministic fingerprint S with $|S| < pn/\alpha$ and a container associated with it with few edges. It is clear that, by design, $I \subset C$.

This proof is a bit magical, we smartly defined this algorithm, now let's keep track of the probability of the event that it is tracking throughout. The probability bounds we get from this event will yield properties (i) and (iii) from [3.9].

Suppose then, that given I, the algorithm queries the vtxs $v_1, \dots, v_M$. We want to know that, given that we chose V_p randomly from the independent sets, what is the probability we got our fingerprint. To that end, define the events

$$\mathcal{E}_0 = \{V_p \in \mathcal{I}(G)\} \tag{3}$$

$$\mathcal{E}_i = \mathcal{E}_{i-1} \cap \begin{cases} \{v_i \in V_p\} & \text{if } v_i \in S \\ \{v_i \notin V_p\} & \text{if } v_i \notin S \end{cases} \tag{4}$$

Smartly calculating $\mathbb{P}(\mathcal{E}_i)$ will yield all properties. But first we must see this event in a way more recognizable by our algorithm. Let S^i and C^i represent the fingerprint and the container after querying the ith vtx,

Claim 3.10.

$$\mathcal{E}_i = \{V_p \in \mathcal{I}(G[C^i])\} \cap \{V_p \subset C^i\} \cap \{S^i \subset V_p\}$$

Proof. To prove the inclusion (lhs to rhs), it suffices to notice that, during the algorithm, we remove precisely only vtxs that got denied and neighbors of vtxs that got accepted into the fingerprint (so if $V_p \in \mathcal{I}(G)$, it must be contained in C^i, and therefore independent in $G[C^i]$). Also it is clear that $\mathcal{E}_i \subset \{S^i \subset V_p\}$. The other inclusion is basically the same. □

In particular,

$$\mathbb{P}(\mathcal{E}_i) = \mathbb{P}(v_i \in V_p \wedge \mathcal{E}_{i-1}) = \mathbb{P}(v_i \in V_p | \mathcal{E}_{i-1}) \mathbb{P}(\mathcal{E}_{i-1})$$

And that motivates one more claim (to be used in the algorithm)

Claim 3.11.

$$\mathbb{P}(v_i \in V_p | \mathcal{E}_{i-1}) = \mathbb{P}(v_i \in C_p^{i-1} | C_p^{i-1} \in \mathcal{I}(G)) < (1-\alpha)p\mathbb{P}(\mathcal{E}_{i-1})$$

where here v_i is such that $\mathbb{P}(v \in C_p^{i-1} \,|\, C_p^{i-1} \in \mathcal{I}(G)) < (1-\alpha)p$

Note that if this were true, by the way we choose the vtxs in the main loop, we would find that $\mathbb{P}(\mathcal{E}_M) < (1-\alpha)^{|S|} p^{|S|}$

Proof of Claim. We want to show

$$\mathbb{P}(v_i \in V_p | V_p \in \mathcal{I}(G[C^{i-1}]) \wedge V_p \subset C^{i-1} \wedge S^{i-1} \subset V_p) = \mathbb{P}(v_i \in C_p^{i-1} | C_p^{i-1} \in \mathcal{I}(G[C^{i-1}]))$$

The last equality is clear from the fact that we are sampling only from C, the first one follows from the fact that conditioning on $\{V_p \subset C^{i-1}\}$ and $\{V_p \in \mathcal{I}(G)\}$, the event $\{S^{i-1} \subset V_p\}$ is independent from $\{v_i \in V_p\}$ (as $S^{i-1} \cup \{v_i\}$ is an indep. set). □

This already gives the first property!

Claim 3.12. (Property (i) of [3.9]) At the end of the algorithm, the fingerprint produced has size at most pn/α.

Proof. Suppose as before, that given I, we queried the vtxs $v_1, \ldots, v_M$ and the algorithm stopped producing a fingerprint S. We note that

$$p^{|S|}(1-p)^{n-|S|} = \mathbb{P}(V_p = S) \leqslant \mathbb{P}(\mathcal{E}_M) \leqslant (1-\alpha)^{|S|}p^{|S|}.$$

In particular, approximating $(1-p)$ by e^{-p} and $(1-\alpha)$ by $e^{-\alpha}$, we find that

$$e^{p(n-|S|)} \leqslant e^{-\alpha|S|}$$

so that $|S| \leqslant pn/\alpha$. □

Now for the second and a bit harder part. We need to show our containers are sparse.

Claim 3.13. When the algorithm ends, $e(G[C]) \leqslant 4\alpha|C|/p$, furthermore $\Delta(G[C]) \leqslant 8\alpha|C|/p$.

Proof. Suppose for sake of contradiction that $e(G[C]) > 4\alpha|C|/p$ and $\mathbb{P}(w \in C_p | C_p \in \mathcal{I}(G)) > (1-\alpha)p$ for every $w \in C$. Take some vtx v with $|N(v) \cap C| \geqslant 8\alpha|C|/p$, note that

$$\mathbb{P}(v \in C_p | C_p \in \mathcal{I}(G)) = \frac{\mathbb{P}(C_p \in \mathcal{I}(G) \wedge C_p \cap N(v) = \varnothing \wedge v \in C_p)}{\mathbb{P}(C_p \in \mathcal{I}(G))} \tag{5}$$

$$= \frac{\mathbb{P}(v \in C_p | C_p \cap N(v) = \varnothing \wedge C_p \in \mathcal{I}(G))\mathbb{P}(C_p \cap N(v) = \varnothing \wedge C_p \in \mathcal{I}(G))}{\mathbb{P}(C_p \in \mathcal{I}(G))} \tag{6}$$

$$= p \cdot \mathbb{P}(C_p \cap N(v) = \varnothing | C_p \in \mathcal{I}(G)) \tag{7}$$

Now we use Chebyshev to upperbound the last probability and get our contradiction. Let $Y = |C_p \cap N(v)|$, we have

$$\mathbb{P}(Y = 0 | C_p \in \mathcal{I}(G)) \leqslant \frac{\mathrm{Var}[Y | C_p \in \mathcal{I}(G)]}{\mathbb{E}[Y | C_p \in \mathcal{I}(G)]^2}.$$

By linearity of expectation,

$$\mathbb{E}[Y|C_p \in \mathcal{I}(G)] = \sum_{u \in N(v)} \mathbb{P}(u \in C_p | C_p \in \mathcal{I}(G)) > 8\alpha(1-\alpha)|C|.$$

Furthermore

$$\mathrm{Var}[Y|C_p \in \mathcal{I}] = \sum_{u,w \in N(v)} \mathbb{P}(u, w \in C_p | C_p \in \mathcal{I}(G)) - \mathbb{P}(u \in C_p | C_p \in \mathcal{I}(G))\mathbb{P}(w \in C_p | C_p \in \mathcal{I}(G))$$

noww, if $u \neq w$, we get (using harris on the first term and the hypothesis on the second)

$$\mathbb{P}(u, w \in C_p | C_p \in \mathcal{I}(G)) - \mathbb{P}(u \in C_p | C_p \in \mathcal{I}(G))\mathbb{P}(w \in C_p | C_p \in \mathcal{I}(G)) \leqslant p^2 - (1-\alpha)^2 p^2 = \alpha^2 p^2$$

for $u = w$, we just get

$$\mathbb{P}(u \in C_p | C_p \in \mathcal{I}(G)) - \mathbb{P}(u \in C_p | C_p \in \mathcal{I}(G))^2 \leqslant p - (1-\alpha)^2 p^2$$

So, in the end $\mathrm{Var}[Y|C_p \in \mathcal{I}(G)] \leqslant 8\alpha|C| + 128\alpha^4|C|^2$. In particular,

$$\mathbb{P}(C_p \cap N(v) = \varnothing | C_p \in \mathcal{I}(G)) \leqslant \frac{8\alpha|C| + 128\alpha^4|C|^2}{64\alpha^2|C|^2} = \frac{1}{8\alpha|C|} + 2\alpha^3$$

if α is smaller than $1/8$ it is clear already that

$$\mathbb{P}(C_p \cap N(v) = \varnothing | C_p \in \mathcal{I}(G)) < (1-\alpha)$$

which together with [5] yields the contradiction.

□

3.4.2 Containers without uniformity imply containers with uniformity

Marcelo's great idea is to prove a stronger form of containers, one that implies [3.3] but furthermore makes no assumptions on the regularity of the Hypergraph. In this section we will state the general form and prove this implication. The proof itself is astounishingly beautiful and leaves everyone shocked ☺. It follows the strategy of [3.9].

Now our containers won't be containers anymore... To each $I \in \mathcal{I}$ and its fingerprint S, there will be an associated a hypergraph $\mathcal{G} = f(S)$ such that I is an independent set of $\mathcal{G}$. Somehow we want to simplify our independent sets to be independent in relation to these $\mathcal{G}$.

Theorem 3.14. Let $\mathcal{H}$ be a hypergraph with $V(\mathcal{H}) = V$ and $|V| = n$. Then for $0 < p < \alpha < 1/8$, there exists $g : \mathcal{I}(\mathcal{H}) \to \mathcal{S} \subset 2^V$ and $f : \mathcal{S} \to 2^{2^V}$, where here we think of $\mathcal{S}$ as the set of fingerprints and for $S \in \mathcal{S}$, $f(S)$ is some hypergraph, such that

1. $\forall S \in \mathcal{S}$, S is small, that is $|S| \leqslant pn/\alpha$.

2. $\mathcal{I}(\mathcal{H}) = \bigcup_{S\in\mathcal{S}} \mathcal{I}(f(S))$ and furthermore $\forall I \in \mathcal{I}(\mathcal{H})$, $I \in \mathcal{I}(f \circ g(I))$.

3. And the strange property for every $I \in \mathcal{I}$, if $\mathcal{G} = f \circ g(I)$, then for every $L \subset I$, we have that
$$\mathbb{P}(L \subset V_p | V_p \in \mathcal{I}(\mathcal{G})) \geqslant (1-\alpha)^{|L|} p^{|L|}. \tag{8}$$

This super general theorem makes absolutely no assumption on the regularity, but to use it, we need to be smart enough to translate (8) into useful properties. The proof itself of it is probably already magical, now we will glimpse how it is used to prove our old container friends.

More precisely, we will want to show the following theorem

Theorem 3.15. Given $\mathcal{H}$ an r uniform hypergraph on n vtxs and $0 < p < 1$, there exists a fingerprint set $\mathcal{S} \subset 2^V$ and an associated containers set $\mathcal{C} \subset 2^V$ (and functions ...) satisfying that

1. $\forall S \in \mathcal{S}, |S| = O(rpn)$.

2. $\forall I \in \mathcal{I}(\mathcal{H})$, there exists $C = C(S(I))$ s.t $I \subset C$.

3. For every $C \in \mathcal{C}$, there exists a cover $\mathcal{G}$ of $\mathcal{H}$, such that every edge $E \in \mathcal{G}$ has size bigger or equal than 2 and
$$w_{p/r}(\mathcal{G}[C]) = \sum_{e\in\mathcal{G}[C]} \leqslant \frac{2p|C|}{r}$$

It is easy to see that for $r = 2$ this is precisely theorem 3.9.

Proof that 3.14 implies 3.15. Fix $\alpha = 1/8r$ and apply 3.14. We can take the same fingerprints $\mathcal{S}$ and, given $\mathcal{G} = f(S)$, we set our containers to be
$$C(S) = \{v \in V : \{v\} \notin \mathcal{G}\}.$$
Properties (1) and (2) are immediate, we are left with proving property (3).

Let $\mathcal{G}$ for $C(S)$ be precisely $f(S)$. Suppose for the sake of contradiction that (3) holds in 3.14 but that $w_{p/r}(\mathcal{G}[C]) \geqslant \frac{2p|C|}{p}$. We will find a set $L \in \mathcal{I}(\mathcal{H})$ that contradicts our assumption, i.e it does not hold that
$$\mathbb{P}(L \subset V_p | V_p \in \mathcal{I}(\mathcal{G})) \geqslant (1 - \frac{1}{8r})^{|L|} p^{|L|}.$$
Opening up this expression as before, we see that
$$\mathbb{P}(L \subset V_p | V_p \in \mathcal{I}(\mathcal{G})) = \frac{\mathbb{P}(L \subset V_p \wedge V_p \in \mathcal{I}(\mathcal{G}))}{\mathbb{P}(V_p \in \mathcal{I}(\mathcal{G}))} = p^{|L|} \frac{\mathbb{P}(V_p \in \mathcal{I}(\mathcal{G}) | L \subset V_p)}{\mathbb{P}(V_p \in \mathcal{I}(\mathcal{G}))}. \tag{9}$$

Because we are conditioning on $L \subset V_p$, we get

$$\{V_p \in \mathcal{I}(\mathcal{G}) | L \subset V_p\} = \{V_p \in \mathcal{I} \wedge V_p \in \mathcal{I}(\partial_L \mathcal{G}) | L \subset V_p\}$$

where $\partial_L \mathcal{G}$ is the link of L in $\mathcal{G}$,

$$\partial_L \mathcal{G} = \{E \setminus L : E \in \mathcal{G}, E \cap L \neq \varnothing\}.$$

So it follows that (using Harris inequality)

$$\mathbb{P}(V_p \in \mathcal{I}(\mathcal{G}) | L \subset V_p) = \mathbb{P}(V_p \in \mathcal{I}(\mathcal{G}) \wedge V_p \in \mathcal{I}(\partial_L \mathcal{G}) | L \subset V_p) \leqslant \mathbb{P}(V_p \in \mathcal{I}(\mathcal{G}) \wedge V_p \in \mathcal{I}(\partial_L \mathcal{G})).$$

Plugging it back in (9), we get

$$\mathbb{P}(L \subset V_p | V_p \in \mathcal{I}(\mathcal{G})) \leqslant p^{|L|} \mathbb{P}(V_p \in \mathcal{I}(\partial_L \mathcal{G}) | V_p \in \mathcal{I}(\mathcal{G})).$$

So to get our contradiction, we need to show following our hypothesis that there exists some $L \in \mathcal{I}(\mathcal{H}[C])$, such that

$$\mathbb{P}(V_p \in \mathcal{I}(\partial_L \mathcal{G}) | V_p \in \mathcal{I}(\mathcal{G})) > \left(1 - \frac{1}{8r}\right)^{|L|}.$$

To make life easier, and use up the r uniformity of $\mathcal{H}$, we will actually show something stronger. We let $\mathcal{G}^{(s)}$ be the edges of size s in $\mathcal{G}$. Clearly $\mathcal{G} = \bigcup_{s=1}^{r} \mathcal{G}^{(s)}$. We will prove that, for some s,

$$\mathbb{P}(V_p \in \mathcal{I}(\partial_L \mathcal{G}^{(s)}) | V_p \in \mathcal{I}(\mathcal{G})) > \left(1 - \frac{1}{8r}\right)^{|L|}.$$

We will use second moment to prove this, just like in the proof of [3.9]! Let $\tilde{V}_p = V_p | V_p \in \mathcal{I}(\mathcal{G})$ (Remember here that we really only care about vertices in C, since $\tilde{V}_p$ won't contain singletons). Just as before, define the random variable

$$X(L, s) = e(\partial_L \mathcal{G}^{(s)}[\tilde{V}_p])$$

and note that

$$\mathbb{P}(V_p \in \mathcal{I}(\partial_L \mathcal{G}^{(s)}) | V_p \in \mathcal{I}(\mathcal{G})) = \mathbb{P}(X(L, s) = 0).$$

So it suffices to lowerbound this probability and we will use Chebyshev for that (If somehow we had a better bound, we could even improve the container lemma!). Let's compute $\mathbb{E}[X(L, s)]$ and $\mathrm{Var}(X(L, s))$.

$$\mathbb{E}[X(L, s)] = \sum_{E \in \partial_L \mathcal{G}^{(s)}} \mathbb{P}(E \subset \tilde{V}_p).$$

Since we can always choose our cover $\mathcal{G}$ to be minimal, that is, there are no edges contained inside others we have that, it follows that each $E \in \partial_L \mathcal{G}^{(s)}$ is an independet set in $\mathcal{G}$. So, letting $\ell = |L|$,

$$\sum_{E \in \partial_L \mathcal{G}^{(s)}} \mathbb{P}(E \subset \tilde{V}_p) \geqslant \sum_{\substack{F \in \mathcal{G}^{(s)} \\ L \subset F}} (1-\alpha)^{s-|L|} p^{s-|L|} = \Delta_{s-\ell}(L)(1-\alpha)^{s-\ell} p^{s-\ell}$$

onde $\ell = |L|$. Now for the variance,

$$\mathrm{Var}[X(L,s)] = \sum_{T,T' \in \partial_L \mathcal{G}^{(s)}} \mathbb{P}(T \cup T' \subset \tilde{V}_p) - \mathbb{P}(T \subset \tilde{V}_p)\mathbb{P}(T' \subset \tilde{V}_p).$$

By the assumption, we can bound

$$\mathbb{P}(T \subset \tilde{V}_p)\mathbb{P}(T' \subset \tilde{V}_p) \geqslant (1-\alpha)^{2(\ell-t)} p^{2(\ell-t)}$$

and, using Harris again,

$$\mathbb{P}(T \cup T' \subset \tilde{V}_p) \leqslant p^{|T \cup T'|}.$$

So we get

$$\mathrm{Var}[X(L,s)] \leqslant \sum_{T,T' \in \partial_L \mathcal{G}^{(s)}} p^{|T \cup T'|} - (1-\alpha)^{2(\ell-t)} p^{2(\ell-t)}.$$

We can now separate in the case that T and T' are disjoint or not. We get

$$\mathrm{Var}[X(L,s)] = \sum_{T \cap T' \neq \varnothing} p^{|T \cup T'|} \quad + \quad (1-\alpha)^{2(\ell-t)} p^{2(\ell-t)}$$

Note how similar the first term is to the Δ in Janson's inequality it showed up here! If

□

Complete this proof! It is very similar to the one in Graph containers

4 Random Graph Theory

4.1 Chromatic number of Gnp

The following is a very nice theorem by Béla!

Theorem 4.1 (Béla 87)**.**

$$\chi(G(n,1/2)) = (1/2 + o(1))\frac{n}{\log_2(n)} \tag{10}$$

The lowerbound follows easily from the fact that $\alpha(G(n,1/2)) \leqslant 2\log_2(n)$ whp [2.3] and that for any n-vtx graph G, $\chi(G) \geqslant n/\alpha(G)$.

The upperbound is a very nice application of Janson's [B.1]. We will just grab independent sets of size $(2-\varepsilon)\log_2(n)$ while we can!.

As well, prove 3.14 , which is very simple, exactly the

Lemma 4.2. For $\varepsilon > 0$, with very high probability, every set of vertices of $G(n, 1/2)$ with size $m \geqslant n/(\log n)^2$, has an independent set of size $(2-\varepsilon)\log_2 n$.

Proof. We will do an union bound over m sets. For that, set $k = (2-\varepsilon)$ and fix a specific set M of size m, note that

$$\mathbb{P}(\alpha(G(n,1/2)[M]) < k) = \mathbb{P}(\alpha(G(m,1/2) < k)).$$

Now, what is $\mathbb{P}(\alpha(G(m,1/2) < k))$? Let $S \in \binom{[m]}{k}$ and denote $B_S = \{S \text{ is indep.}\}$. It is clear now that,

$$\mathbb{P}(\alpha(G(m,1/2)) < k) = \mathbb{P}\left(\bigcap_S B_S^c\right)$$

and we are in the hypothesis of Janson. Note that

$$\mu = \sum_{S \in \binom{[m]}{k}} \mathbb{P}(S \text{ is indep.}) = \binom{m}{k}\left(\frac{1}{2}\right)^{\binom{k}{2}} \geqslant \left(\frac{m}{k}2^{-k/2}\right)^k \gg (n^{\varepsilon})^{\log n}$$

and

$$\Delta = \sum_{\substack{S,T \in \binom{[m]}{k} \\ S \cap T > 2}} \mathbb{P}(S \cup T \text{ indep.}) = \sum_{l=2}^{k-1} \binom{m}{k}\binom{k}{l}\binom{m-k}{k-l}\left(\frac{1}{2}\right)^{2\binom{k}{2}-\binom{l}{2}}.$$

Now it's a bit ugly to see but this thing here is convex and the sum is majorized by the last two elements. We get

$$\begin{aligned}\Delta &\simeq \binom{m}{k}\binom{k}{2}\binom{m-k}{k-2}\left(\frac{1}{2}\right)^{2\binom{k}{2}-1} + \binom{m}{k}k(m-k)\left(\frac{1}{2}\right)^{\binom{k}{2}} \\ &\simeq 2\binom{m}{k}^2 \frac{k^4}{m^2}2^{-2\binom{k}{2}} + km\mu = \frac{2k^4}{m^2}\mu^2 + km\mu \leqslant \frac{3k^4}{n^2}\mu^2.\end{aligned}$$

By the second part of Janson, we get that

$$\mathbb{P}\left(\bigcap_S B_S^c\right) \leqslant \exp(-cn^2/k^4) \lll 2^{-n}$$

So we can easily do a union bound, over all sets. □

Proof of Theorem 4.1. Just use the previous lemma and remove independent sets untill you have less than $n/(\log n)^2$ vertices left. At that point, do a trivial coloring. □

4.1.1 Concentration of the Chromatic number

We will look into a very weird type of result. We will show that with high probability, if p is in the right range, $\chi(G(n,p))$ is super concentrated. Our main objective is to prove the following surprising statement

Theorem 4.3 (Alon-Krivilevich 95)**.** Let $\delta > 0$, $p = n^{-1/2-\delta}$ and $G \sim G(n,p)$ then there exists $t = t(n)$ s.t w.h.p

$$\chi(G) \in \{t, t+1\}.$$

Remark 4.4. This is a beautiful theorem, although an extremely weird one, since we don't even know the value of t.

From now on, we will always think of $G \sim G(n,p)$, to diminish notation clustering. The main tool for this kind of result is Luczak's lemma. It showed up in the first proofs of statements of this kind.

Lemma 4.5 (Luczak)**.** For any $\varepsilon > 0$, let $t = t(n,p,\varepsilon)$ be the least integer s.t

$$\mathbb{P}(\chi(G) \leqslant t) \geqslant \varepsilon.$$

Let X be a r.v that counts the minimal number of vertices to be removed from G to make it t-colorable. Then, if λ is such that $\varepsilon = e^{-\lambda^2/2}$, we have

$$\mathbb{P}(X > 2\lambda\sqrt{n}) \leqslant \varepsilon.$$

Proof. We will do a clever martingale argument (note crucially here that we don't really know that t is). X satisfies the Lipchitz condition for the vertex exposure martingale, so that it is typically concentrated [A.6]. More precisely,

$$\mathbb{P}(X \leqslant \mathbb{E}X - \lambda\sqrt{n}) < e^{-\lambda^2/2} \quad \text{and} \quad \mathbb{P}(X \geqslant \mathbb{E}X + \lambda\sqrt{n}) < e^{-\lambda^2/2}. \tag{11}$$

In particular, by the definition of X, and t, we know that with probability at least ε, $X = 0$. By the first equation, this means that $\mathbb{E}X < \lambda\sqrt{n}$. Plugging that into the second equation we get the result. □

Let's get ready for theorem [4.3]. We will need some properties of $G(n,p)$ at this range.

Lemma 4.6. [Properties of $G(n,p)$] Let $\delta > 0$ and $G \sim G(n,p)$ with $p = n^{-1/2-\delta}$. The following properties all hold with high probability (less than some ε for big n).

(i) $\Delta(G) \leqslant 3d = 3np$.

(ii) $\chi(G) \geqslant d/2\log n$.

(iii) For some fixed $r = r(\delta)$ we have that for every $U \subset V(G)$ with $|U| \leqslant O(\sqrt{n})$, $e(G[U]) \leqslant r|U|$. Note that, by degeneracy, this makes every set $W \subset V(G)$ with $|W| \leqslant O(\sqrt{n})$ $2r$-colorable.

(iv) If $\delta > 1/5$, then for every set $U \subset V(G)$ with $|U| \leqslant O(n^{1-\delta})$, $e(G[U]) \leqslant |U|n^{1/10}$.

Furthermore, let $X_{u,v}$ be the number of paths of length 3 (edges) between u and v. We also find that

(v) If $\delta \geqslant 1/6$, we find that $\max_{u,v \in V(G)} X_{u,v} \leqslant \log n$.

(vi) For $0 < \delta < 1/6$, this maximum is at most $d^3 \log n/n$.

Proof. All of these follow from the first moment. (i) is just Chernoff, (ii) follows directly from [2.3]. (iii) and (iv) are as well simple first moment bounds. The only interesting ones are (v) and (vi). To show them we note that, fixed $u, v \in V(G)$, every edge participates in at most $c_0 = 3/\delta$ paths of length 3 between u and v w. h.p. In particular, each path of length 3 between u, v touches at most $3c_0$ others. So there must be $X_{u,v}/(3c_0 + 1)$ disjoint paths between u and v. Using this bound, we show the result. □

Now we are ready to prove the theorem. It suffices to show the following proposition.

Proposition 4.7. Suppose $G(\delta)$ is a graph on n vertices satisfying the conclusions of the previous lemma [4.6]. Moreover, suppose, following lemma [4.5], that $G \geqslant t \geqslant d/2 \log n$ and that G has a set $U_0 \subset V(G)$ s.t $|U| \leqslant 2\lambda\sqrt{n} = O(\sqrt{n})$ and $\chi(G[V(G) \setminus U]) = t$. Then G is $t + 1$ colorable.

Remark 4.8. We will prove a totally deterministic statement, using probabilistic methods, to prove a probabilistic result! The irony.

I will lay out the proof plan. The idea is to find a set U with $U_0 \subset U$, $|U| = O(\sqrt{n})$ such that every vertex in $V(G) \setminus U$ has few neighbors in U (say $10r$ which is constant). We know that $V(G) - U$ is t colorable and that U, by its size, is $2r$ colorable as well. Fix any t-coloring of $V(G) \setminus U$. Say $U = \{u_1, \ldots, u_k\}$, set $N(u_i) = N_G(u_i) \setminus U$, also set $N(U) = \bigcup_{j=1}^{k} N(u_j)$. We will try to exchange $2r$ colors from each neighborhood $N(u_i)$ with a new color $\{t + 1\}$ as to leave $2r$ possible color choices for the vertex u_i, after that we can just use degeneracy to greedily color U with $2r$ colors and finish the proof (Having properly (t+1)-colored our graph).

Although the strategy is the same, how we choose the colors will be different depending on δ. So for now I will only show the $\delta < 1/5$ case.

Proof of 4.7. (For $\delta \leqslant 1/5$) Let's formalize the previous paragraph. Fix r constant as in [4.6]. To find U, start with $U = U_0$ of size $2\lambda\sqrt{n}$ and while there are vertices v s.t. $d_U(v) > 10r$,

add them to U. We cannot have that $|U| \geqslant 2|U_0|$ as that would break the hypothesis that $e(G[U]) \leqslant r|U|$. So we really have $|U| = O(\sqrt{n})$ and every vertex outside of U sends in at most constant edges.

Now let $f : V(G) \setminus U \to \{1, \dots, t\}$ be a proper coloring of the rest of the world. To help us recolor the neighborhood of U, we will do a unneeded but very helpful step of defining an auxiliary graph H of $N(U)$. For each $1 \leqslant i \leqslant k$, let W_i be a copy of $N(u_i)$, we define H with vertex set $\bigsqcup_{i=1}^{k} W_i$. Note that each vertex $a \in W$ corresponds to a vertex a' in $N(U)$, we connect a to b if $a' \sim b'$. Note that f induces a proper coloring f' of H as well (in some sense we are just blowing up the vertices of $N(U)$ according to how many neighborhoods they belong to). Also, a set A in H is independent if and only if the corresponding set is independent in $N(U)$. Very importantly, note that each edge of U corresponds to at most $(10r)^2$ edges in H.

Having defined this graph we can be more precise about our plan of attack. Each W_i is colored with t colors. For each $1 \leqslant i \leqslant k$ we want to choose $2r$ colors from $\{1, \dots, t\}$, making a set C_i, such that $W_0 = \bigcup_{j=1}^{k} \{w \in W_j : f'(w) \in C_j\}$ is independent. If we do so, we can just paint W_0 with the color $t + 1$, paint the corresponding vertices in $N(U)$ with this color, and leave $2r$ colors available for each u_i.

We will choose this C_i randomly (but differently depending on δ). If $\delta \leqslant 1/5$, we will use the LLL [2.20], if $\delta > 1/5$ we will do a clever counting argument. As we are in the $\delta \leqslant 1/5$ case, let's proceed.

For each i choose $C_i \subset \binom{[t]}{2r}$ uniformly and randomly. For each edge $e = \{w_1, w_2\}$ in H, with $w_1 \in W_{i_1}$ and $w_2 \in W_{i_2}$, define the bad event B_e to be $\{f'(w_1) \in C_{i_1} \text{ and } f'(w_2) \in C_{i_2}\}$. Note that for any edge e,

$$\mathbb{P}(B_e) \leqslant \left(\frac{2r}{t}\right)^2 = O((\log n/d)^2).$$

What about our dependency graph? We note that B_{w_1,w_2} is completely independent of B_{w_3,w_4} if $\{W_{i_1}, W_{i_2}\} \cap \{W_{i_3}, W_{i_4}\} = \varnothing$. That is, by revealing C_j for $j \notin \{i_1, i_2\}$ we gain no information about $B_{\{w_1,w_2\}}$. So what is the degree of B_e? It is the number of edges that touch W_{i_1} or W_{i_2}. Because $|W_{i_1}| = N(u_i) \leqslant 3d = o(\sqrt{n})$, $e(W_i) = O(d)$. What about edges between W_{i_1} and W_j? Because of the assumption over $X(u_{i_1}, u_j)$ (number of paths of length 3 between both vtxs), and that each edge of G is corresponds to at most $(10r)^2$ edges in H, we note that

$$e(W_{i_1}, W_j) \leqslant \begin{cases} (10r)^2 \log n & \text{if } \delta > 1/6 \\ (10r)^2 d^3 \log n/n & \text{if } \delta \leqslant 1/6 \end{cases}$$

Overall, this means that

$$d(B_e) \leqslant \begin{cases} O(d) + k \cdot O(\log n) = O(n^{1/2} \log n) & \text{if } \delta > 1/6 \\ O(d) + k \cdot O(d^3 \log n/n) = O(d^3 \log n/n^{1/2}) & \text{if } \delta \leqslant 1/6 \end{cases}$$

In both cases, as $\delta \leqslant 1/5$, $d \geqslant n^{2/5}$, we get that

$$\mathbb{P}(B_e) \cdot d(B_e) = o(1)$$

and by the LLL [2.20] we get our result. □

Now we delve into what should be the easier case (closer to what people already had solved) of $\delta > 1/5$. In our computations we will need to use that the result was already proven for $\delta > 3/8$, which is a one page proof by Luczak [https://link.springer.com/article/10.1007/BF01205080]. It is worth reading it, it has some similar ideas.

Our main goal will be the same, still choose for each W_i a set of colors C_i, such that the set W_0 as defined before is an independent set. But here we will have to do some alteration, we will take each C_i in a way that C_j and C_i are compatible for $j < i$, furthermore, we will take each C_i a bit bigger so that we can remove the colors of the edges inside W_i, which are few. Let's engage the proof now.

Proof of 4.7. (For $1/5 < \delta \leqslant 3/8$) As before define U, H and its induced coloring f' with $t \geqslant d/2 \log n$ colors. By Lemma 4.6, for each $s \leqslant k$, the sets $N(u_{i_1}), \dots, N(u_{i_s})$ have size less than $s3d \leqslant n^{1-\delta}$, and so span less than $3sdn^{1/10}$ edges. In particular, we can always order the u_i's in such way that that $N(u_i)$ sends at most $3dn^{1/10}$ edges for all $i' < i$. It follows that, in this order,

$$e\left(W_i, \bigcup_{j<i} W_j\right) \leqslant 3(10r)^2 dn^{1/10} = O(dn^{1/10})$$

as well. For each i, we will choose a random $2r + \log n$ set of colors from the ones that are "available". A color $c \in \{1, \dots, t\}$ is not available if there exists an $j < i$ and an edge (w_j, w_i) between W_j, W_i with $f'(w_j) = c$. Call x_i the probability that for some $i' \leqslant i$, less than $t/2$ colors are available when choosing $C_{i'}$, we will try to show that $x_k < 1$. Obviously, $x_1 = 0$, now we note that, because $\delta \geqslant 1/6$, $e(W_j, W_i) \leqslant O(\log n)$, so each color chosen in C_j may kill at most $\log n$ colors when choosing C_i. Because we need to kill $t/2$ colors, we get that

$$\begin{aligned} x_i &= x_{i-1} + (1 - x_{i-1}) \binom{O(dn^{1/10})}{\frac{t}{2^{O(\log n)}}} \left(\frac{2r + \log n}{t/2}\right)^{\frac{t}{2^{O(\log n)}}} \\ &\leqslant x_{i-1} + e^{-n^{1/10}} \end{aligned}$$

So clearly, $x_k \leqslant \sqrt{n} e^{-n^{1/10}} \to 0$ (in this calculation it seems we use $\delta \leqslant 3/8$ somewhere). Now, this means we can choose the colors C_i such that they are compatible. Inside each W_i there are at most $\log n$ edges, kill the colors corresponding to them, we are still left with $2r$ colors for each C_i that make W_0 an independent set. The proof follows the same way as before now. □

4.2 Thresholds for bounded degree spanning subgraphs

In this section, we will prove a marvelous theorem by Alon and Furëdi that gives upperbounds on the critical probability of spanning subgraphs.

Theorem 4.9 (Alon-Furëdi)**.** Let $H \subset K_n$ be a graph with $\Delta(H) \leqslant \Delta$ bounded (not depending on n), if

$$p \gg \left(\frac{\log n}{n}\right)^{1/\Delta}$$

then

$$\mathbb{P}(H \subset G(n,p)) \approx 1.$$

Remark 4.10. This theorem is useful for H being hamiltonian cycles or triangle factors, or really any spanning subgraph! I've written approximately 1 because in truth I wanted to say when n tends to infinity, but we can't do that as the graph H is not fixed (we would have to consider a sequence of graphs).

It has a very smart proof. We will divide H into big independent classes and, using Hall's theorem, iteratively embed these classes into parts of $G(n,p)$. To embed such a part, we construct an auxiliary bipartite graph (which will be random with independent edges) and show that it satisfies Hall's condition with very high probability.

For our strategy to work, we need a nice little lemma.

Lemma 4.11. Let H be a n-vtx graph with max degree $\Delta = \Delta(H)$. Then we can split $V(H)$ into 2Δ independent parts $V_1, V_2, \dots V_{2\Delta}$ such that each part is big, i.e $|V_i| \geqslant \lfloor n/4\Delta \rfloor$ for every i.

Proof. Separate empty sets $V_1, \dots V_{2\Delta}$ and sequentially add each vertex v of H to a part of minimal size which it is allowed, that is $N(v) \cap V_i = \varnothing$. Suppose for sake of contradiction that in the end, we were left with a part with less than $n/4\Delta$ elements.

Suppose this class is W, note that every one of the non-neighbors of W could possibly have been added to it, but they weren't because they had a even smaller class to join. But there are ate least $n - \Delta|W| \geqslant 3n/4$ such elements. Each was added to a class of size less than $n/4\Delta$, but there are at most 2Δ such classes, so we could have at most $n/2$ such elements, absurd! □

Let's give continuity to the proof of the theorem.

Proof of 4.9. Apply the previous lemma to H^2 to find independent classes $V_1, \dots V_{\Delta^2}$ such that for every i, $|V_i| \geqslant n/4\Delta^2$ and, crucially, every element the distance between elements $u, w \in V_j$ is at least 3 (this is important when definining the auxiliary graph).

Now fix a partition $W_1, \dots, W_{\Delta^2}$ of $[n]$ with $|W_i| = |V_i|$ and consider $G(n,p)$ over it. We will embed each part V_i of H into the corresponding W_i sequentially. So our objective is to find a bijection $f_i : V_i \to W_i$, that respects neighbors of previous choices. That is, given $v \in V_i$, suppose it has a neighbor $u \in V_j$ with $j < i$, it must be that $\{f_i(v), f_j(u)\}$ is an edge in $G(n,p)$. To this end, embed V_1 into W_1 arbitrarily, and for each $i > 1$ and $v \in V_i$, define the allowed images of v to be the set

$$A(v) = \{w \in W_i : \forall i < j, \forall u \in N(v) \cap V_j, \{f_j(u), w\} \in G(n,p)\}.$$

We may consider this an auxiliary bipartite graph F on (V_i, W_i), where the neighborhood of each $v \in V_i$ is $A(v)$. Now, notice that for $w \in A(v)$, w must be connected to the previous at most Δ neighbors chosen. So an edge in F occurs with probability at least p^Δ. Note also, that two edges of the form $\{v_1, w_1\}$ and $\{v_2, w_2\}$ are completely independent if $w_1 \neq w_2$ (as they depend on disjoint edges of $G(n,p)$). If $w_1 = w_2$, the edges keep being independent, since there are no paths of length 2 connecting vertices in V_i. Because of how we chose p, with very high probability F satisfies Hall's condition (each edge is present with prob at least $\log n/n$), so we can always find such bijection and we are done! □

I wasn't really nitty picky on proving that Hall's hold with high probability here, as it is not the crux of the proof. But I will do it for completion.

Lemma 4.12. Let (A, B) be a bipartite graph with $n = |A| = |B|$. Suppose that for every set $X \subset A$ we have that $N(X) \geqslant |X|$, then there exists a perfect matching in (A, B).

Proof. We proceed by induction. If $n = 1$, the result follows easily. For bigger n, call a set $X \subsetneq A$ critical if $|X| = |N(X)|$. If A contains no critical subsets, then pick any edge (x, y), and notice Hall's condition still is valid for $(A - x, B - y)$ and we can apply induction. Suppose A contains a critical X (pick X maximal), note that as $N(A) = B$, it must be that $B - N(X) \subset N(A - X)$. As Hall's condition is valid both for $(X, N(X))$ and $(A - X, B - N(X))$ (by maximality of X), we can apply induction on both parts. □

Now to finish the proof, we just need to prove the following lemma,

Lemma 4.13. Consider a random bipartite graph $G = (A, B)$ with $|A| = |B| = n$ where each edge (a, b) appears with probability $p = C(\log n/n)$. Then G contains a matching with high probability.

Proof. It suffices to check that G satisfies Hall's condition. Fix a set X of size m, the probability that $|N(X)| \leqslant |X|$ is bounded by

$$\binom{n}{m}(1-p)^{(n-m)m} \leqslant \begin{cases} (\frac{ene^{-p(n-m)}}{m})^m & m \leqslant n/2 \\ (\frac{ene^{-pm}}{n-m})^{n-m} & m \geqslant n/2 \end{cases}$$

So we always get that this probability is bounded by

$$\left(ne^{-pn/2}\right)^{\min(m,n-m)} \ll 1$$

and is summable on m if C is a big enough constant. □

4.3 A proof of the Kahn-Kalai Conjecture

We will show in this subsection a very cool conjecture that gives thresholds for a big family of monotone properties.

The idea is the following. Thinking about $G(n,p)$, what is the threshold for an event $\{H \subset G(n,p)\}$ for a fixed H of constant size? It is easy to show a lower bound by the first moment. If $m(H) = \max\{v(F)/e(F) : F \subset H\}$ and $p \ll n^{-m(H)}$, then

$$\mathbb{P}(H \subset G(n,p)) \leqslant \mathbb{E}\#\{H \subset G(n,p)\} \to 1.$$

By doing Janson or second moment, it is easy to show that this bound is tight!

Notice here that we are actually looking for F, the densest subgraph which is the hardest to find. This threshhold is determined by the threshhold of F! Put it in hypergraph terms, this is because the hypergraph that encodes F copies is a cover for the hypergraph that encodes H copies.

Let's state and prove the famous Kahn-Kalai conjecture!

Theorem 4.14. Let $\mathcal{H}$ be a hypergraph on vertex set V of size n and suppose that for every covering $\mathcal{F}$ of $\mathcal{H}$, $w_p(\mathcal{F}) \geqslant 1$. Then

$$\mathbb{P}(\mathcal{H}[V_{C(\varepsilon)p\log n}] = \varnothing) < \varepsilon.$$

Remark 4.15. Think of $\mathcal{H}$ encoding matchings or hamiltonian cycles! If the first moment threshhold is $1/n$, but the real threshhold is $\log n/n$.

Remark 4.16. This is a very nice theorem, but all applications of it actually use the fractional version. Where we define a fractional covering as a function $\lambda : 2^V \to [0,1]$ s.t for every edge $E \in \mathcal{H}$, $\sum_{S \subset E} \lambda(S) \geqslant 1$.

The main idea is to prove the contrapositive. If the probability of $\mathcal{H}[V_{Cp\log n}]$ being empty is big, then we can find a cheap cover. So let's try that. For now on suppose

$$\mathbb{P}(\mathcal{H}[V_p] = \varnothing) \geqslant 1/2.$$

Now, let's consider $X_1, \ldots, X_s$ each being a q-random set with $q = \alpha p$ where α is a constant and $s = \log n$. We will iteratively build a cover $\mathcal{F}$ of $\mathcal{H}$ which we will show is cheap with positive probability.

Preemptively define $k_i = 2^{-i}n$. Set our first partial cover to be

$$\mathcal{F}_1 = \min\{E \setminus X_1 : E \in \mathcal{H}\}$$

where min here means we are take the minimal set. Clearly $\mathcal{F}_1$ is a covering of $\mathcal{H}$. Now we separate it in two parts,

$$\mathcal{F}_1^{\leqslant k_1} = \{E \in \mathcal{F}_1 : |E| \leqslant k_1\} \quad \& \quad \mathcal{F}_1^{>k_1} = \{E \in \mathcal{F}_1 : |E| > k_1\}.$$

Now we will use X_2 to build a covering of $\mathcal{F}_1^{\leqslant k_1}$ the same way as before. We define

$$\mathcal{F}_2 = \min\{E \setminus X_2 : E \in \mathcal{F}_1^{\leqslant k_1}\}$$

separate the big edges and the small edges and go on. The idea here is that the big edges are cheap, we can't find them in V_p so let's only use them to make our cover. Proceeding this way, setting

$$\mathcal{F}_i = \min\{E \setminus X_i : E \in \mathcal{F}_{i-1}^{\leqslant k_{i-1}}\}$$

and $\mathcal{F}_i^{\leqslant k_i}$, $\mathcal{F}_i^{>k_i}$ accordingly we get, by definition, that

$$\mathcal{F} = \bigcup_{i=1}^{s} \mathcal{F}_i^{>k_i}$$

is a covering for $\mathcal{H}$. We now only have to prove one main lemma:

Lemma 4.17. For some big enough C.

$$\mathbb{E}[w_p(\mathcal{F}_i^{>k_i})] \leqslant C^{-k_i}$$

Let's see how it implies the theorem. Assuming it, and considering the random covering $\mathcal{F}$, we have that

$$\mathbb{E}w_p(\mathcal{F}) \leqslant \sum_{i=1}^{s} \mathbb{E}w_p(\mathcal{F}_i^{>k_i}) \leqslant \sum_{i=1}^{s} C^{-k_i} = O(C^{-1}).$$

Now, by assumption $\mathbb{P}(\mathcal{H}[V_p] = \varnothing) > 1/2$ so with $1/2$ chance, no $\mathcal{F}_i$ is empty and furthermore with probability at least $1 - O(C^{-1})$, $w_p(\mathcal{F}) < 1$. So there exists a non-empty cheap covering of $\mathcal{H}$.

Proof of the lemma. The proof will be a smart counting and will use the minimality. Let's looks closely at $\mathbb{E}w_p(\mathcal{F}_i^{>k_i})$. We may separate it on the possible value of X_i, for this consider $\mathcal{G}(I) = \min\{E - I : E \in \mathcal{F}_{i-1}^{\leqslant k_{i-1}}\}$ to be the next covering if we chose X_i as I. Then it is clear that

$$\mathbb{E}w_p(\mathcal{F}_i^{>k_i}) = \sum_{I \subset 2^V} \mathbb{P}(X_i = I) w_p(\mathcal{G}^{>k_i}(I)).$$

Now let's condition on $\mathcal{F}_{i-1}$ and open up this right-hand side. Note that

$$w_p(\mathcal{G}^{>k_i}(I)) = \sum_{k>k_i} p^k e(\mathcal{G}^k(I)).$$

Now we set $A_{m,k} = \{(I,J) : |I| = m, J \in \mathcal{G}^k(I)\}$. We just proved that the previous expectation is exactly

$$\sum_{m\leqslant n}\sum_{k>k_i} p^k q^m (1-q)^{n-m}|A_{m,k}|.$$

Now comes the ingenius trick! Let's bound the possible size of $|A_{m,k}|$ by considering a simple map. Consider $g : A_{m,k} \to \binom{V}{m+k}$ that sends $g(I,J) \to I \cup J$. Let's see that the pre-image of any value of this thing cannot be too big! So how many guys can be such that $g(I,J) = T$ where T is some set? Note first that by definition, if $(I,J) \in A_{m,k}$, then $I \cup J = T$ contains an edge $\mathcal{F}_{i-1}^{\leqslant k_{i-1}}[T]$. Actually, furthermore, if $E \in \mathcal{F}_{i-1}^{\leqslant k_{i-1}}[T]$, then it must be that $J \subset E$. Note that if this wasn't the case, then $J \cap E$ would be a smaller set and it would contradict minimality. So we actually have that for all F satisfying that, it must be that $F \subset \cap_{E\in\mathcal{F}_{i-1}^{\leqslant k_{i-1}}[T]} E$. In particular, the number of such F is bounded by $2^{k_{i-1}}$. We find that, for $k \geqslant k_i$,

$$|A_{m,k}| \leqslant 2^{k_{i-1}}\binom{n}{m+k} \leqslant 2^{2k}\binom{n}{m+k}. \tag{12}$$

Plugging this back in the equation, we get that it is less than

$$\sum_{m\leqslant n}\sum_{k>k_i} p^k q^m (1-q)^{n-m} 2^{2k}\binom{n}{m+k}.$$

Swapping the sums and adding a k factor to q^m and removing one from $(1-q)^{n-m}$ we get that this is at most

$$\sum_{k>k_i} \frac{2^{2k}p^k(1-q)^k}{q^k}\sum_{m\leqslant n} q^{m+k}(1-q)^{n-m-k}\binom{n}{m+k}.$$

But notoriously, this inner sum is always less than 1, so we get

$$\mathbb{E}[w_p(\mathcal{F}_i^{>k_i})] \leqslant \sum_{k>k_i}\frac{(4p(1-q))^k}{q^k} = O((\alpha/4)^{-k}).$$

So by taking α big enough we get our result!. □

Remark 4.18. (About the Theorem) The same proof works under the assumption that $\mathcal{H}$ is r-uniform. In this case, we define $k^i = 2^{-i}r$ and get a $\log r$ instead of a $\log n$ in the statement of the main theorem.

4.3.1 Sunflowers

Maybe this should be in the extremal set theory section!

We will see an application of the previous theorem in this subsection. It actually came before and was the missing piece that people used to prove the Kahn-Kalai conjecture. But thankfully, Kahn-Kalai implies it ☺.

Definition 4.19. Given a r uniform hypergraph $\mathcal{H}$, a t-sunflower of $\mathcal{H}$ is a collection of edges $S_1, \dots, S_t \in \mathcal{H}$ such that for all $i \neq j$, $S_i \cap S_j = \bigcap_{i=1}^t S_i$.

Remark 4.20. We call the big intersection the core of the sunflower and we denote a petal to be an edge without the core.

A very nice extremal theorem by Erdös and Rado from the 60's states that every big enough $\mathcal{H}$ has a t-sunflower. Note that we don't even care about the vertices of $\mathcal{H}$.

Theorem 4.21. If $\mathcal{H}$ is r-uniform and $|\mathcal{H}| \geqslant (rt)^r$, then $\mathcal{H}$ contains a t-sunflower.

Proof. Let's begin by noticing that if $\Delta(\mathcal{H})$ was bounded by Δ and $|HH|$ was big enough, then we could just do it greedily and find a sunflower with empty core. More precisely, we could choose a set S_1, remove every edge that intersects it - at most $r\Delta$ - and pick up another edge. So, if $(t-1)r\Delta \leqslant |HH|$ we are done!

So at first we will do exactly that. We would be very happy if this was the case, but we can't garantee it. What if some vertex belonged to a whole lot of edges of $|HH|$? Well in that case we would just add it to the core. Motivated by that define L to be a maximal set such that

$$d_{\mathcal{H}}(L) \geqslant p^{|L|}|\mathcal{H}|$$

for any $p > |\mathcal{H}|^{-1/r}$. Note that $\varnothing$ easily satisfies this, so this maximal exists. Furthermore, $|L| < r$, because if it wasn't the case, then $|L| = r$, $d_{\mathcal{H}}(L) \leqslant 1$, but $p^{|L|}|\mathcal{H}| > 1$. The big idea is now taking L as the core. Note that, if we define $\mathcal{H}_L = \{E - L : E \in \mathcal{H}\}$, by assumption, if $v \notin L$, then

$$d_{\mathcal{H}_L}(v) \leqslant d_{HH}(L \cup v) < p^{|L|+1}|\mathcal{H}| \leqslant p|\mathcal{H}_L|.$$

So $\Delta(\mathcal{H}_L) < p|\mathcal{H}_L|$ meaning we can find a matching in $\mathcal{H}_L$ of size at least

$$\frac{|\mathcal{H}_L|}{r\Delta(\mathcal{H}_L)} \geqslant \frac{1}{pr} = |\mathcal{H}|^{1/r}/r.$$

In particular, if $(tr)^r \leqslant |\mathcal{H}|$, we can find a t-sunflower. □

Remark 4.22. This proof is very nice and we encapsulate all we need in this smart definition of L. But actually, L gives a whole lot more of information. It gives information of all codegrees as well! Note that if $T \cap L = \varnothing$, then $d_{\mathcal{H}_L}(T) < p^{|L|+|T|}|\mathcal{H}| \leqslant p^{|T|}|\mathcal{H}_L|$. The next good idea is to use this information as well.

Definition 4.23. We say that a hypergraph $\mathcal{F}$ is p-spread if for every $S \subset V(\mathcal{F})$ we have that

$$d_{\mathcal{F}}(S) \leqslant p^{|S|}|\mathcal{F}|.$$

Note that crucially, as we defined, $\mathcal{H}_L$ is p-spread with $p = |HH|^{-1/r}$. Now, this is the crazy part, we will use the Kahn-Kalai Conjecture to find a matching more efficiently in $\mathcal{H}_L$. For this we need a nice result that says that p spread implies no cheap covering.

Lemma 4.24. Let $\mathcal{F}$ be a p-spread hypergraph. Then for all fractional covers $\lambda : 2^V \to [0,1]$ of $\mathcal{F}$, we have $w_p(\lambda) \geqslant 1$.

Proof. Marcelo's proof of this uses linear programming duality smartly but there is a simpler way. Let λ be a covering for $\mathcal{F}$, then note that

$$w_p(\lambda) = \sum_{S \subset V} p^{|S|}\lambda(S) = \sum_{S \subset V} p^{|S|}\lambda(S)\frac{1}{d_{\mathcal{F}}(S)}\sum_{E \in \mathcal{F}} \mathbf{1}[S \subset E].$$

Swapping the summation order, we get

$$\sum_{E \in \mathcal{F}}\sum_{S \subset V} \frac{p^{|S|}\lambda(S)}{d_{\mathcal{F}}(S)}\mathbf{1}[S \subset E].$$

Now by the p-spreadness hypothesis, we know $d_{\mathcal{F}}(S) \leqslant p^{|S|}|\mathcal{F}|$. Then we can lowerbound the last expression by

$$\sum_{E \in \mathcal{F}} \frac{1}{|\mathcal{F}|}\sum_{S \subset V} \lambda(S)\mathbf{1}[S \subset E] \geqslant \sum_{E \in \mathcal{F}} \frac{1}{|\mathcal{F}|} \geqslant 1$$

where we used crucially that λ is a covering of $\mathcal{F}$. □

Now we can find a new bound for sunflowers!

Theorem 4.25. There exists an absolute C such that if $\mathcal{H}$ is an r uniform hypergraph and $(C \log rt \log t)^r < |\mathcal{H}|$, then $\mathcal{H}$ contains a t sunflower.

Proof. Define $p > |\mathcal{H}|^{-1/r}$, L and $\mathcal{H}_L$ exactly as before. We noted that $\mathcal{H}_L$ is p-spread, and by the previous lemma, $\mathcal{H}_L$ has no cheap covers. Now, if C is the constant in Park-Pham's theorem, setting $M = C \log r$, we have that

$$\mathbb{P}(\mathcal{H}_L[V_{Mp}] = \varnothing) < 1/2.$$

So by sprinkling

$$\mathbb{P}(\mathcal{H}_L[V_{lMp}] = \varnothing) < \frac{1}{2^l}.$$

Consider a random t-coloring of the vertices of $\mathcal{H}_L$. That is set $V = X_1 \cup X_2 \cdots \cup X_t$, where each X_i is a $1/t$ random set when looked at alone. If $\mathbb{P}(\mathcal{H}_L[X_i] = \varnothing) < 1/t$ then, by the union bound

$$\mathbb{P}(\exists i \mathcal{H}_L[X_i] = \varnothing) < 1$$

in particular, with positive probability, $\mathcal{H}_L[X_i]$ is not empty for all i. So we found a matching in $\mathcal{H}_L$. What bounds did we get? Well, if $1/2^l < Mpl$, then we can find a $1/Mpl$ matching. Fixing $l = \log t$ and asking $1/Mpl > t$, we see that if

$$\frac{|\mathcal{H}|^{1/r}}{C \log r \log t} = t$$

or in order terms, if

$$|\mathcal{H}| > (C \log rt \log t)^r$$

we can find a t-sunflower. □

4.3.2 A proof of one of Talagrand's Conjectures

Following the Kahn-Kalai Conjecture, which was already proven for fractional coverings. Talagrand made a very natural question:

Question 4.26. Let $\mathcal{H}$ be a hypergraph with a cheap fractional cover λ. Does there exist a cheap cover $\mathcal{G}$ of $\mathcal{H}$?

This was tackled incompletely by Pham in a recent paper [2025] and most recently (last week as of writing this) A.I settled it.

Theorem 4.27 (Open A.I)**.** Let $\mathcal{H}$ be a hypergraph containg a fractional covering λ with $w_p(\lambda) < 1$ and $\lambda(\varnothing) = 0$. For an absolute constant $C > 0$, there exists a integral cover $\mathcal{G}$ of $\mathcal{H}$ such that $w_{Cp}(\mathcal{G}) < 1$.

The proof looks a lot like Pham's, but it has some very weird robot steps. We will make some definitions, replace the sprinkling steps in the Kahn-Kalai proof to some random coloring, and try to do a smarter switching step. By switching here I mean the nice proof of inequality [12].

Definition 4.28. Fix $\mathcal{H}$ and a fractional covering λ. For each edge $E \in \mathcal{H}$, we define a vertex measure $\sigma_E : E \to [0, 1]$ given by

$$\sigma_E(x) = \frac{1}{M_E} \sum_{S \subset E : x \in S} \frac{\lambda(S)}{|S|}.$$

where M_E is simply $\sum_{S \subset E} \lambda(S)$ so that σ_E sums up to 1.

Now, the idea is that we should really prioritize vertices with high σ_E when making up our cover. Things will seem to go sideway but let's follow along the proof.

The key lemma looks a lot like the key lemma from Kahn-Kalai, it states that if some probability is sufficiently big, then we can build up a cheap integral cover. In our previous case, this probability was $\mathbb{P}(\mathcal{H}[V_{Cp\log n}] = \varnothing)$.

Lemma 4.29. Let $\mathcal{H}$ be a hypergraph, $p \in (0,1)$ and λ a fractional covering of $\mathcal{H}$ with $\lambda(\varnothing) = 0$. For a big constant D, consider s such that

$$\sum_{j=1}^{s} D^j p = 1.$$

Let $X_1, \dots X_s$ be a random coloring of V, such that independently for each $x \in V$, $\mathbb{P}(x \in X_i) = D^i p$. Suppose furthermore that $w_{Dp}(\lambda) < 2^{-10}$. Define $X_{>i} = \bigcup_{j>i} X_j$. Let $\mathcal{E}$ be the event that for every $E \in \mathcal{H}$, there exists $1 \leqslant i \leqslant s$, such that

$$\sum_{x \in X_{>i}} \sigma_E(x) > 2^{-i}.$$

If $\mathbb{P}(\mathcal{E}) > 1/2$ then $\mathcal{H}$ contains a cover $\mathcal{G}$ with $w_{Cp}(\mathcal{G}) < 1$.

This weird lemma will give us the result. Let us show first that it is sufficient.

Lemma 4.30. If $\mathcal{H}$ admits a covering λ with $w_{Dp}(\lambda) < 2^{-10}$, then for $\mathcal{E}$ defined before, $\mathbb{P}(\mathcal{E}) > 1/2$.

This proof is a bit magical, really something out from the machine. We will define a smart random variable Z which has low expectation, but takes big values when $\mathcal{E}$ does not happen. So $\mathbb{P}(\mathcal{E})$ must be low. So from now on we will be interested in the event that there exists an edge E that is tracked, i.e for every $1 \leqslant i \leqslant s$,

$$\sum_{x \in X_{>i}} \sigma_E(x) \leqslant 2^{-i}.$$

Let $a : V \to [s]$ be the coloring function defined by $a(x) = i$ such that $x \in X_i$. Note that if E is tracked, then by a simple summation,

$$\sum_{x \in E} \sigma_E(x) a(x) \leqslant 2.$$

Furthermore, define for a set S the average color of S to be $\bar{a}(S) = \frac{1}{|S|} \sum_{x \in S} a(x)$. Finally (and this is a crucial point) note that if we consider the measure on the subsets S of E given by $\lambda(S)/M_E$. Then

$$\sum_{S \subset E} \frac{\lambda(S)}{M_E} \bar{a}(S) = \sum_{S \subset E} \sum_{x \in S} \frac{\lambda(S)}{M_E |S|} a(x)$$

which after swapping the summation is precisely

$$\sum_{x \in E} a(x)\sigma_E(x) \leqslant 2.$$

Now after all of this discussion, let's prove the previous lemma.

Proof of 4.30. Fix C (which we will determine later) and define the magic random variable Z given by

$$Z = \sum_{S \subset V} \lambda(S) C^{4-\bar{a}(S)}.$$

Note crucially that if the event $\mathcal{E}$ does not happen, that is some edge E is tracked. Then

$$Z \geqslant \sum_{S \subset E} \lambda(S) C^{4-\bar{a}(S)}.$$

But we have seen that, if E is tracked,

$$\sum_{S \subset E} \frac{\lambda(S)}{M_E} \bar{a}(S) \leqslant 2.$$

So, by Markov, for at least half the sets (under the $\lambda(.)/M_E$ measure) S we have that $\bar{a}(S) \leqslant 4$. We conclude that

$$Z \geqslant M_E \sum_{S \subset E} \frac{\lambda(S)}{M_E} C^{4-\bar{a}(S)} \geqslant \sum_{S \subset E : \bar{a}(S) \leqslant 4} \frac{\lambda(S)}{M_E} \geqslant \frac{1}{2}$$

But let's compute the expectation of Z. We have that

$$\mathbb{E}Z = C^4 \sum_{S \subset V} \lambda(S) \mathbb{E} C^{-\bar{a}(S)}.$$

Now comes the nice bit, because of how we have chosen our coloring, we may use independence to open up $\mathbb{E}[C^{-\bar{S}}]$. It becomes

$$\prod_{x \in S} \mathbb{E} C^{-a(x)/|S|} = (\mathbb{E} C^{-a(x)/|S|})^{|S|}.$$

as all $a(x)$ are i.i.d. But $a(x)$ has a simple, calculable distribution, we have that

$$\mathbb{E} C^{-a(x)/|S|} = \sum_{j=1}^{s} C^{-j} p D^j \approx \frac{pD}{C^{1/|S|}}.$$

as such,

$$\mathbb{E}Z \leqslant C^4 \sum_{S \subset V} \lambda(S) \frac{p^{|S|} D^{|S|}}{C} \leqslant C^3 w_{Dp}(\lambda) \leqslant C^3 2^{-10}.$$

If C is not too big, we are fine and can do the Markov argument. Note howerver that for this to work we need some assumptions on D and p, we just need that p is smaller than some small constant.

So

$$C^3 2^{-10} \geqslant \mathbb{E}[Z] \geqslant \frac{1}{2}(1 - \mathbb{P}(\mathcal{E}))$$

so $\mathbb{P}(\mathcal{E}) \geqslant 1/2$, by choosing an appropriate C. □

Now let's try to prove the main lemma using a switching argument similar to those seen before in 12.

the following lemma doesnt really make sense, we have to make it correct following the random variables

Lemma 4.31. Let $\mathcal{H}$ be a hypergraph with a fractional covering λ ($\lambda(\varnothing) = 0$) with $w_{Dp}(\lambda) < 2^{-10}$. For some big constant D, set X_i as before. If for every edge $E \in \mathcal{H}$, E is not tracked, i.e there exists i such that

$$\sum_{x \in X_{>i}} \sigma_E(x) > 2^{-i}$$

then there exists a cheap integral covering $\mathcal{G}$ with $w_{Cp}(\mathcal{G}) < 1$ for some big absolute C.

5 Fun Additive Results

In this section we will see some fun results that concern or permeate additive combinatorics. Some of them may not be additive in nature but use tools common in the additive world, so they may appear.

5.1 The Combinatorial Nullstellensatz

In this maybe wrongly located section we will explore nice applications of a powerful theorem by Alon. The combinatorial Nullstellensatz allows us to find non-zeros of certain polynomials with specific degree condtions. We will use it very much like in Section [1], devising smart polynomials to get bounds on some nice problems.

Theorem 5.1 (Alon's Combinatorial Nullstellensatz)**.** Let $f \in \mathbb{F}[X_1, \ldots, X_n]$ be a n variable polynomial of degree $d = t_1 + \cdots + t_n$, suppose furthermore the coefficient of $X_1^{t_1} \ldots X_n^{t_n}$ is non zero. If there are sets $S_1, \ldots S_n \subset \mathbb{F}$ with $|S_i| > t_i$, then there exists $X \in S_1 \times \cdots \times S_n$ with $f(X) \neq 0$.

Proof. We proceed by induction! For $n = 1$ the result is trivial, every polynomial f of degree d contains at most d roots. Suppose it is valid for all numbers less or equal than $n - 1$, let's prove it for n. Let f be given as in the assumption and consider the polynomial on one variable

$$g_n(X_n) = \prod_{s \in S_n} (X_n - s) = X_N^{|S_n|} + \ldots$$

where the dots are lower order terms. Apply long division to f with g_n to find h and r s.t

$$f(X_1, \dots, X_n) = h(X_1, \dots, X_n)g_n(X_n) + r(X_1, \dots, X_n)$$

where crucially, $d(h) \leqslant d - |S_N|$ and the maximum degree of X_n in r is at most $|S_n| - 1$. So

$$r(X_1, \dots, X_n) = \sum_{j=0}^{|S_n|-1} X_n^j r_j(X_1, \dots, X_{n-1}).$$

Now, we know that the degree of $X_1^{t_1} \dots X_n^{t_n}$ of this thing is non zero. But, because $|S_n| > t_n$, it is clear that we can expand hg_n as

$$hg_n = hX_n^{|S_n|} + O(X_n^{d-1}).$$

So hg_n contains no monomial of type $X^{\boldsymbol{t}}$. In particular, it must be that r_{t_n} has a non zero coefficient for $X_1^{t_1} \dots X_{n-1}^{t_{n-1}}$ and degree exactly d. So we can apply induction to it to find a non-zero root $x_1, \dots x_{n-1}$. Fixing this and considering $f(x_1, \dots, x_{n-1}, X_N)$ as a one variable polynomial ad applying the $n = 1$ cases gives the result. □

Now let's see some applications of this thing. All of them are similar in the sense that we construct a smart polynomial and apply this theorem, yielding a contradiction on the degree hypothesis.

Theorem 5.2 (Cauchy-Davenport)**.** Let $A, B \subset \mathbb{Z}_p$, then $|A + B| \geqslant \min(|A| + |B| - 1, p)$.

Proof. If $|A|+|B| > p$, then it is clear that $A \cap (x - B) \neq \varnothing$ for every $x \in \mathbb{Z}_p$, so $A + B = \mathbb{Z}_p$. Now suppose for sake of contradiction that $|A + B| < |A| + |B| - 1 \leqslant p - 1$, consider the polynomial

$$f(X, Y) = \prod_{z \in A+B} (X + Y - z).$$

Note that $d(f) = |A + B| < |A| + |B| - 1 \leqslant p - 1$ by hypothesis. Furthermore, consider $t_1 + t_2 = |A + B|$ with $t_1 \leqslant |A| - 1$ and $t_2 \leqslant |B| - 1$, and the coefficient of the monomial $X^{t_1}Y^{t_2}$ which is $\binom{|A+B|}{t_1} \neq 0 \pmod p$. So, by the Nullstellensatz [5.1], there exists $x \in A$, $y \in B$ s.t $f(x, y) \neq 0$ which is absurd. □

After that we saw a fun application regarding covering the hypercube with hyperplanes. Note that it is trivial to cover $Q_n = \{0, 1\}^n$ with two hyperplanes $\{x_1 = 0\} \cup \{x_1 = 1\}$. What can we say if we want to avoid an element, say 0? We certainly can do it with n, as $Q_n \setminus \{0\} \subset \bigcup_{j \leqslant n} \{x_j = 1\}$. This is surprisingly the best.

Theorem 5.3. Let $H_1, \dots H_m$ be hyperplanes such that $0 \notin H_i$ for any i and furthermore they cover the rest of the hypercube, $Q_n \subset \bigcup_{i=1}^m H_i$. Then $m \geqslant n$.

Proof. Again we proceed by contradiction. Suppose there was a covering with $n-1$ hyperplanes H_i, we will cook up a smart polynomial to apply Noga's theorem. We want a f that is 0 whenever $\vec{x} \in H_i$ and is 0 at $\vec{0}$, because then we could apply our theorem and get a contradiction. So consider the equations that define the hyperplanes H_i as

$$H_i = \{X \in \{0,1\}^n : A_i^T X - b_i = 0\}$$

for vectors A_i and constants b_i. Crucially, as $\vec{0} \notin H_i$, $b_i \neq 0$ for all i. Consider now the following function (on $\mathbb{R}$ even)

$$f(X) = \prod_{i=0}^{m}(A_i^T X - b_i) - (1 - \mathbf{1}[X = \vec{0}]) \prod_{i=1}^{m}(-b_i).$$

This function satisfies what we want but its not a polynomial. We can make it easily be one, by remembering $1 - \mathbf{1}[X = \vec{0}] = \prod_{j=1}^{n}(1 - X_j)$. Putting this in our equation we get,

$$f(X) = \prod_{i=0}^{m}(A_i^T X - b_i) - \prod_{i=1}^{m}(-b_i) \prod_{j=1}^{n}(1 - X_j).$$

which is, by assumption, a degree n polynomial. Taking $t_i = 1$ and $S_i = \{0,1\}$ we get by the Nullstellensatz a non-zero, which is different than $\vec{0}$, absurd! □

The last application we saw was a seemingly very weird theorem of Alon, Friedman and Kalai. It states that a graph that is very very close to being regular, has a regular subgraph. A very nice corollary of this, that Sebastião is presenting eventually, is the extremal number of d-regular graphs. Using this, Pyber proved that if $e(G) \geqslant C(d)n \log n$, G contains a d-regular graph. Let's see the AFK (cool name) theorem.

Theorem 5.4 (AFK)**.** Let p be a prime and suppose G is a graph satisfying

$$d(G) > 2p - 2 \quad \& \quad \Delta(G) \leqslant 2p - 1$$

where here $d(G)$ is the average degree. Then G contains a p-regular subgraph.

Proof. Because we know that we will use the Nullstellensatz, we have to cook up a good polynomial f. As we are looking for general subgraphs (not induced), it seems a good idea for our variables to be X_e taking values $\{0,1\}$ where e is an edge. We want the degree of every vertex to be either 0 (the vertex doesn't connect to anyone and we can take it out) or p. By taking values mod p, we can make use of Fermat's Little Theorem, to write an indicator for that. Crucially, in $\mathbb{Z}_p$, $\mathbf{1}[x \neq p = 0] = 1 - x^{p-1}$. So let's consider the product

$$\prod_{u \in V} 1 - \Big(\sum_{e=\{u,v\} \in E} X_e \Big)^{p-1}$$

Note that this product is not 0 on $H \subset G$ if and only if every vertex of H has degree in $\{0, p\}$. But there is a problem, we might take $H = \varnothing$, so just like the previous application, we have to remove that possibility. We may consider

$$f(X) = \prod_{u \in V} 1 - \left(\sum_{e=\{u,v\} \in E} X_e \right)^{p-1} - \prod_{e \in E} (1 - X_e).$$

Now what is the degree of f? We clearly have $e(G)$ on the second term and at most $(p-1)n$ on the first. By the average degree condition, $e(G) > (2p-2)n/2 = (p-1)n$, so the second term wins! Now, considering $t_e = 1$ and $S_e = \{0, 1\}$, we can apply the theorem to get a non zero which corresponds to a non-empty subgraph $H \subset G$ such that for every $v \in H$, $d(v) = 0 \pmod{p}$. As $\Delta(G) \leqslant 2p-1$, it must be that $d(v) \in \{0, p\}$ exactly as we wanted to show! □

Now let's see a cool application of this theorem (which seems surprisingly weak).

Theorem 5.5 (Pyber 85)**.** Let G be a graph on n vertices with $e(G) \geqslant C(d) n \log n$. Then G contains a d-regular subgraph.

Proof. The idea of the proof is to find a very close to regular bipartite subgraph H and apply the previous theorem. So the first step is clean G up, we first take a bipartition with half the edges and then consider a subgraph of this bipartition with $\delta(H) \geqslant d(G)/4 \approx C(d) \log n$. This can be done by removing low degree vertices. Now suppose our graph H has parts (A, B) with $n \geqslant |A| \geqslant |B|$ and that minimum degree, we will remove edges until every vertex on A has degree exactly δ. Now, consider A_1 a minimal set inside A such that $|N(A_1)| \leqslant |A_1|$, note that A_1 satisfies Hall automatically (and furthermore $|A_1| = |N(A_1)|$). So we can apply Hall and find a matching M_1 between (A_1, B_1). Now notice that, if we separate M_1, out, A_1 becomes $\delta - 1$ regular. So we can iterate this procedure. This yields a sequence $(A_1, B_1, M_1), (A_2, B_2, M_2), \dots (A_\delta, B_\delta, M_\delta)$. We will show we can take a union of some of these matchings and apply the AFK theorem. To that end, fix $2p$ a prime constant way bigger than d. As $|A_\delta| \geqslant 1$, and we are doing $\delta = C(d) \log n$ steps, we see that there must be some p of these steps in such way that the ratio of the sizes between $|A_m|$ and $|A_{m+p}|$ is small. To prove this, note that

$$n \geqslant \prod_{i=1}^{\delta} |A_i|/|A_{i+1}|$$

so

$$\log n \geqslant \sum_{i=1}^{\delta} \log(|A_i|) - \log(|A_{i+1}|)$$

in particular, there is m such that $\log |A_i| - \log |A_{i+p}| \leqslant p \log(n)/\delta$. By making C big enough, $|A_i|/|A_{i+p}| \leqslant (1 + \varepsilon)$ where ε we can take as small as we want depending only on

C. By considering H on (A_{i+p}, B_i) with edges $M_i \cup M_{i+1} \cdots \cup M_{i+p}$, we can apply AFK and conclude. □

A (Sub)Exponential Tailbounds

I just took these from my notes in random matrices.

Lemma A.1. (Hoeffding's Lemma) Let X be a scalar r.v taking values in $[a, b]$. Then for any $t > 0$

$$\mathbb{E}e^{tX} \leqslant e^{t\mathbb{E}X}(1 + O(t^2\sigma^2 \exp(O(t(b-a))))). \tag{13}$$

In particular

$$\mathbb{E}e^{tX} \leqslant e^{t\mathbb{E}X} \exp(O(t^2(b-a)^2)). \tag{14}$$

Proof. Normalize X as to $\mathbb{E}(X) = 0$ and $b - a = 1$. And consider the Taylor expansion of e^{tX},

$$e^{tX} = 1 + tX + O(t^2X^2 \exp(O(t)))$$

taking expectation yields the first result. The second one follows from assymptotics of the exponential function. □

Theorem A.2. (McDiamird's Theorem) Let $(R_i)_{i=1}^n$ be a tuple of ranges and $X = (X_i)_{i=1}^n$ a random vector taking values in their product. Let $F : \prod_{i=1}^n R_i \to \mathbb{R}$ be a coordinate-wise bounded function (this is a strong Lipchitz condition). That is changing any coordinate x_i may change at most c_i the total value. Then

$$\mathbb{P}(|F(X) - \mathbb{E}F(X)| > \lambda\sigma) \leqslant C \exp(-c\lambda^2)$$

for some $C, c > 0$ and $\sigma^2 = \sum_{i=1}^n c_i^2$.

Proof. By simmetry, we may only consider $\mathbb{P}(F(X) - \mathbb{E}F(X) > \lambda\sigma)$. We turn to Hoeffding's Lemma [A.1], consider

$$\mathbb{E}\exp(tF(X)) = \mathbb{E}\big[\mathbb{E}[\exp(tF(X))|X_1, \dots X_{n-1}]\big],$$

we should try to understand $\mathbb{E}[\exp(tF(X))|X_1, \dots X_{n-1}]$. We have that

$$\mathbb{E}[\exp(tF(X))|X_1, \dots X_{n-1}] = \mathbb{E}[\exp(t(F(X) - \mathbb{E}_{X_n}F(X))) \exp(t\mathbb{E}F(X))|X_1, \dots X_{n-1}]$$

applying (14) to the first product, (here we are conditioning on $X_1, \dots X_{n-1}$)

$$\mathbb{E}[\exp(t(F(X) - \mathbb{E}_{X_n}F(X))) \exp(t\mathbb{E}F(X))|-] \leqslant \exp(O(t^2c_n^2))\mathbb{E}[\exp(tF(X))|-]$$

and we can iteratively apply the theorem to the new coordinate-wise bounded function

$$G(X_1, \dots X_{n-1}) = \mathbb{E}[F(X)|X_1, \dots X_{n-1}].$$

Notice G has the same coefficients as F, because

$$\begin{aligned} |G(y, x_2 \dots, x_{n-1}) - G(z, x_2 \dots, x_{n-1})| &= \left| \int_{R_n} F(y, \dots, x_n) dx_n - \int_{R_n} F(z, \dots, x_n) dx_n \right| \\ &= \left| \int_{R_n} F(y, \dots, x_n) - F(z, \dots, x_n) dx_n \right| < c_1, \end{aligned}$$

and here we can take any other coordinate instead of the first. □

Corollary A.3. (Hoeffding's inequality) Let $(X_i)_{i=1}^n$ be real r.v's taking value in bounded intervals $[a_i, b_i]$. Then, if $S = \sum_{i \leqslant n} X_i$, we have

$$\mathbb{P}(|S - \mathbb{E}S| > \lambda\sigma) < C \exp(-c\lambda^2)$$

for some $C, c > 0$, where $\sigma^2 = \sum_{i \leqslant n} (b_i - a_i)^2$.

Proof. Apply [A.2] with $F(X_1, \dots X_n) = X_1 + \cdots + X_n$. □

Theorem A.4. (Chernoff) Let $(X_i)_{i=1}^n$ be independent scalar random variables with $|X_i| < K$ a.s, with mean μ_i, and variance σ_i^2. Then for any $\lambda > 0$, one has

$$\mathbb{P}(|S_n - \mu| \geqslant \lambda\sigma) \leqslant C \exp(-c \min(\lambda^2, \frac{\lambda\sigma}{K}))$$

for some absolute $C, c > 0$, where $S_n = \sum_{i=1}^n X_i$, $\mu = \sum_{i=1}^n \mu_i$ and $\sigma^2 = \sum_{i=1}^n \sigma_i^2$.

Proof. The proof is again the use of the exponential moment with lemma [A.1]. Suppose we normalize each variable so that $K \leqslant 1$ and $\mu_i = 0$, then $\mu = 0$ and for any $0 \leqslant t \leqslant 1$,

$$\mathbb{E}(\exp(tS_n)) = \prod_{i=1}^n \mathbb{E}(\exp(tX_i)) = \prod_{i=1}^n (1 + O(t^2 \sigma_i^2 \exp(O(t)))).$$

using the usual $1 + x \leqslant e^x$ ineq, we find

$$\mathbb{E}(\exp(tS_n)) \leqslant \exp(t\mathbb{E}(S_n)) \exp(O(t^2) \exp(O(t))\sigma^2) = \exp(O(t^2\sigma^2)),$$

now we use Markov and optimize on t following the restraint on $0 \leqslant t \leqslant 1$. (Suppose the hidden constant is D) we have that

$$\mathbb{P}(S_n \geqslant \lambda\sigma) \leqslant \exp(Dt^2\sigma^2 - t\lambda\sigma)$$

which yields the result after optimizing t following the constraints. The other inequality is equivalent.

□

Corollary A.5. (Tailbounds for the Binomial distribution) Let $B(n,p)$ be the binomial distribution. We have that

$$\mathbb{P}(|B(n,p) - np| \geqslant \delta np) \leqslant C\exp(-c\delta^2 np)$$

for some absolute C, c.

Proof. Note that

$$\mathbb{P}(|B(n,p) - np| \geqslant \lambda\sqrt{np}) \leqslant \mathbb{P}(|B(n,p) - np| \geqslant \lambda\sqrt{np(1-p)}) \leqslant C\exp(-c\lambda^2)$$

by A.4. Applying it with $\lambda = \delta\sqrt{np}$ yields the result. □

Theorem A.6. (Azuma's Inequality) Let $X_1, X_2, \ldots X_n$ be a sequence scalar r.v's with $|X_i| \leqslant 1$ almost surely. Assume also that we have the martingale difference property

$$\mathbb{E}(X_i|X_1, \ldots, X_{i-1}) = 0$$

almost surely for all $i = 1, \ldots, n$. Then for any $\lambda > 0$, the sum $S_n := X_1 + \cdots + X_n$ satisfies

$$\mathbb{P}(|S_n| \geqslant \lambda\sqrt{n}) \leqslant C\exp(-c\lambda^2)$$

for some absolute $C, c > 0$.

Proof. We prove this by calculating exponential moments and using Hoeffding's Lemma again. Note that $S_n = S_{n-1} + X_n$ so that

$$\mathbb{E}(e^{tS_n}) = \mathbb{E}(e^{tS_{n-1}}e^{tX_n}).$$

By taking the expectation of conditioning on the first $n-1$ values, we get that

$$\mathbb{E}(e^{tS_n}) = \mathbb{E}\mathbb{E}(e^{tS_{n-1}}e^{X_n}|X_1, \ldots, X_{n-1}) = \mathbb{E}[e^{tS_{n-1}}\mathbb{E}(e^{tX_n}|X_1, \ldots, X_{n-1})]$$

Now, by A.1, we have that $\mathbb{E}(e^{tX_n}|X_1, \ldots, X_{n-1}) \leqslant e^{O(t^2)}$, so by iterating we get the result. □

Theorem A.7 (Subexponential Bernstein inequality)**.** Let $X_1, \ldots, X_n$ be independent, mean 0, subexponential r.v's. Then, for every $t \geqslant 0$, we have

$$\mathbb{P}\left\{\left|\sum_{i=1}^n X_i\right| \geqslant t\right\} \leqslant 2\exp\left(-c\min\left(\frac{t^2}{\sum_{j=1}^n ||X_j||_{\psi_1}^2}, \frac{t}{\max_j ||X_j||_{\psi_1}}\right)\right)$$

where c is an absolute constant.

Proof. The proof is again only using the exponential Markov method, but as in the proof of Chernoff, we have some limiting on λ as the variables are only subexponential. □

B Poisson Paradigm

Theorem B.1 (Janson Inequalities)**.** Let $A_1, A_2 \dots A_t \subset [N]$ be a family of sets. Consider the events $B_j = \{A_j \subset [N]_p\}$. Then

$$\mathbb{P}\left(\bigcap_{j=1}^{t} B_j^c\right) \leqslant \begin{cases} \exp(-\mu + \Delta/2) \\ \exp(-\mu^2/2\Delta) \qquad \text{if } \mu \leqslant \Delta \end{cases} \tag{15}$$

where

$$\mu = \sum_{i=1}^{t} \mathbb{P}(B_j)$$

and

$$\Delta = \sum_{i \sim j} \mathbb{P}(B_i \cap B_j)$$

(here the sum being over ordered pairs).

Remark B.2. We also have the trivial lowerbound

$$\mathbb{P}(\bigcap_{j=1}^{t} B_j^c) \geqslant \prod_{j=1}^{t} \mathbb{P}(B_j^c).$$

Remark B.3. The following is a version of the previous inequality with more general lowerbounds (and it has a better presentation as well).

Theorem B.4. Let $t_1, \dots, t_n$ be boolean random variables. Let $\mathcal{A}$ be a collection of sets $A \subset [n]$ (we may think of it as hypergraph edges). Consider the random variable (or the random boolean polynomial)

$$X = \sum_{A \in \mathcal{A}} \prod_{j \in A} t_j.$$

We have

$$\mathbb{P}(X \leqslant \mathbb{E}X - T) \leqslant \exp(-T^2/\Delta)$$

where Δ is defined just as before being

$$\Delta = \sum_{A \cap B \neq \varnothing} \mathbb{E} \prod_{j \in A \cup B} t_j.$$

Tao's proof of B.1. Let's see a beautiful proof different from the one in the Probabilistic Method. We will rely as before on the exponential method. Note that, defining $F(t) = \mathbb{E}\exp(-tX)$ and using Markov, we get that

$$\mathbb{P}(X \leqslant \mathbb{E}X - T) = \mathbb{P}(\exp(-tX) \geqslant \exp(-t(\mathbb{E}X - T)) \leqslant F(t)\exp(-t(\mathbb{E}X - T)).$$

So, taking logs, an upper bound on $\log F(t) - t(\mathbb{E}X - T)$ yields an upper bound on our probability. Janson's ingenious trick is noting that $F(0) = 1$ and writing the $\log F(t)$ as an integral, so that

$$\log F(t) - t(\mathbb{E}X - T) = \int_0^t \frac{F'(s)}{F(s)} ds - t(\mathbb{E}X - T).$$

Now, $F'(t) = \mathbb{E}[-X \exp(-tX)]$ so the last expression becomes

$$-\int_0^t \frac{\mathbb{E}[X \exp(-sX)]}{F(s)} ds - t(\mathbb{E}X - T).$$

So flipping signs, we want a lower bound on

$$\int_0^t \frac{\mathbb{E}[X \exp(-sX)]}{F(s)} ds + t(\mathbb{E}X - T)$$

Now we remind that X is of the form $\sum_{A \in \mathcal{A}} \prod_{j \in A} t_j$ and we use linearity inside the the integral to get

$$\sum_{A \in \mathcal{A}} \int_0^t \frac{\mathbb{E}[(\prod_{j \in A} t_j) \exp(-yX)]}{\mathbb{E} \exp(-yX)} dy$$

Letting E_A be the event that for all $j \in A$, $t_j = 1$, we find that

$$\mathbb{E}[(\prod_{j \in A} t_j) \exp(-yX)] = \mathbb{P}(E_A)\mathbb{E}[\exp(-yX)|E_A].$$

Writing $X = Y_A + Z_A$ where

$$Y_A = \sum_{\substack{B \in \mathcal{A} \\ A \cap B \neq \varnothing}} \prod_{j \in B} t_j \quad \text{and} \quad Z_A = \sum_{\substack{B \in \mathcal{A} \\ A \cap B = \varnothing}} \prod_{j \in B} t_j$$

we get that

$$\mathbb{E}[\exp(-yX)|E_A] = \mathbb{E}[\exp(-y(Y_A + Z_A))|E_A] = \mathbb{E}[\exp(-yZ_A)]\mathbb{E}[\exp(-yY_A)|E_A].$$

substituting back in the sum, we get

$$\sum_{A \in \mathcal{A}} \mathbb{P}(E_A) \int_0^t \frac{\mathbb{E}[\exp(-sZ_A)]\mathbb{E}[\exp(-sY_A)|E_A]}{F(s)} ds.$$

Since clearly $Z_A \leqslant X$, we have that $\mathbb{E}[\exp(-sZ_A)] \geqslant F(s)$, from which the independent term cancels. So we get a lowerbound

$$\sum_{A \in \mathcal{A}} \mathbb{P}(E_A) \int_0^t \mathbb{E}[\exp(-sY_A)|E_A] ds$$

Applying Jensen's inequality for $\mathbb{E}[\exp(-sY_A)|E_A] \geqslant \exp(-s\mathbb{E}[Y_A|E_A])$ we get a new lower-bound of

$$\int_0^t \sum_{A\in\mathcal{A}} \mathbb{P}(E_A)\exp(-s\mathbb{E}[Y_A|E_A])ds.$$

We may apply Jensen's again using the fact that $\sum_{A\in\mathcal{A}}\mathbb{P}(E_A) = \mathbb{E}X$ to get that this last integral is lower bounded by

$$\mathbb{E}X\int_0^t \exp\Big(-s\sum_{A\in\mathcal{A}}\frac{\mathbb{P}(E_A)\mathbb{E}[Y_A|E_A]}{\mathbb{E}X}\Big)ds = \mathbb{E}X\int_0^t \exp(-s\frac{\Delta}{\mathbb{E}X})ds$$

we thus conclude by noticing that we can lowerbound the full expression by

$$\frac{(\mathbb{E}X)^2}{\Delta}(1-\exp(-t\frac{\Delta}{\mathbb{E}X})) - t(\mathbb{E}X - T).$$

maximizing over t (or setting it T/Δ) we get the result. □